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BACHELOR THESIS

Physiological and geodata analyses of methane reduction potential in ruminants

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Handed in: [Date]

ABSTRACT

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LIST OF ABBREVIATIONS

6-Aza	6-Azauracil.
8-Aza	8-Azahypoxanthin.
AA	aminoacids.
AB	antibiotic.
CFU	colony forming unit.
CRT	common reporting table.
Cyc-Se.	cycloserin.
DE	digestibility of food.
DEU	Germany.
DMSO	dimethylsulfoxid.
EnterFe	enteric fermentation.
E. coli DH5α	*Escherichia coli DH5*α.
E. coli	*Escherichia coli*.
EtOH	ethanol.
GE	gross energy intake.
LULUCF	land use and land use change and forestry.
M. boviskoreani	*Methanobrevibacter boviskoreani AbM4*.
M.	*Methanobrevibacter*.
M. millerae	*Methanobrevibacter millerae ZA-10*^{T}.
M. thaueri	*Methanobrevibacter thaueri CW*^{T}.
M. wolinii	*Methanobrevibacter wolinii SH*^{T}.
NZL	New Zealand.
OD600	optical density at 600 nm.

Puro	puromycin.
Sqr.	square.
USA	United States of America.
Ym	methane conversionfactor.

1 INTRODUCTION

The following section presents the results regarding a potential mitigation strategy against methane emissions: the immunization of cattle (Baca-González et al., 2020). As explained above, the bachelor's thesis - and thus the results - can be divided into two main sections:

1. The biochemical part, which involves a fundamental analysis of methane-producing archaea isolated from the rumen of cattle, as well as an attempt to develop genetic tools for them.
2. The geographical part, which, based on Buendia et al. (2019) findings, examines how much methane can be reduced and what role this can play in combating climate change.

In the enteric fermentation process of ruminants CH_4 are formed through methanogenic archaea. One discussed mitigation strategy to reduce mathane resulting from cattle farming is to immunisize the cattle against these specific archaea (Malyugina et al., 2025). For developing a specific immunization strategy, the isolated methanogenic archaea strains from cattle rumen has to be well understood. Although some strains have already been isolated from the rumen of cattle, they have hardly been studied to date (Baca-González et al., 2020). This bachelor's thesis aims to address this issue.
In doing so, the focus will primarily be on two strains: *Methanobrevibacter thaueri CW*^{T} (*M. thaueri*) and *Methanobrevibacter boviskoreani AbM4* (*M. boviskoreani*). Besides these two main strains, some experiements were also done on *Methanobrevibacter wolinii SH*^{T} (*M. wolinii*) and *Methanobrevibacter millerae ZA-10*^{T} (*M. millerae*). These two methanogenic archaea were also isolated from the rumen of cattle. Goal of the investigation, that took place, was to detect several basic characteristics and try to implement first genetic engineering tools.

1.1 Biochemestry

Bin mit der Karte so noch unzufrieden, auch wenn sie skizzenhaft, glaube ich schnell zeigt was alles grob bei unseren Konstrukten drauf sein sollte. Bin mir auch nopch unsicher ob ich so eine Art Figure hier in die Einleitung mit reinpacke, bei der Einführung unserer Konjugations/ Transformations technick oder erst im result part -> vmtl hier aber sinnvoller

1.2 Geography

DE has a-non linear relation towards GE (see eq. (4):6).

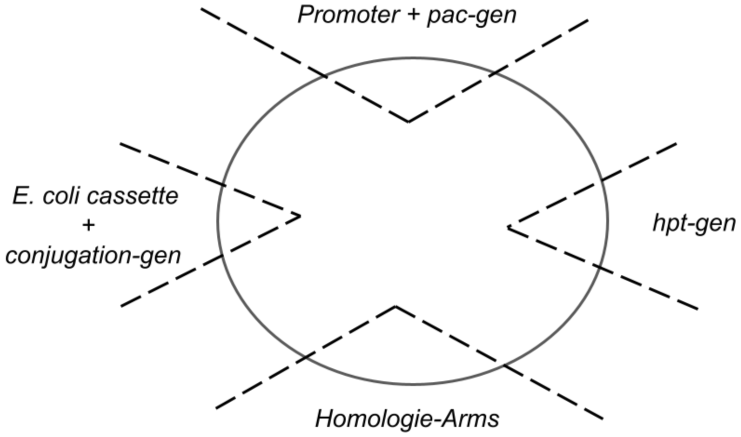

Figure 1.1: Sketch of the general structure of the suicide plasmids for the transformation of *Methanobrevibacter thaueri CW^T* (*M. thaueri*) and *Methanobrevibacter boviskoreani AbM4* (*M. boviskoreani*) via conjugation using *E. coli*. The plasmid contains the *E. coli* cassette with and *catp* as a selection marker using chloramphenicol, as well as the conjugation gene *traJ*. In addition, all plasmids contain the *hpt* gene to enhance counterselection and the two homology arms that flank the genomic *hpt* gene of the corresponding organism. The final important region is the one containing the *pac* gene and its associated promoter, which serves as a selection marker.

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2 DATA AND MATERIALS

2.1 Biochemistry

2.2 Geography

2.2.1 Study area

The geodata analyses of methane reduction potential in rumineszenz will focuson the impact of the immunization of cattle. Two emission sectors of agriculture discussed in the Buendia et al. (2019) could be effected of this. On the one hand methane resulting resulting from enteric fermentation and on the other hand from manure management. However, in this bachelor thesis the manure management sector will not be investigated, primarily due to as-yet-unclear links to a possible microbiome shift, as well as time constraints.

All calculations are based on the methodology described in Buendia et al. (2019) for calculating CH_4 emission from enteric fermentation in cattle.

Furthermore, is the calculation done for only three countries: Germany (DEU), New Zealand (NZL) and United States of America (USA). These three countries were picked, because of three main reasons.

First, the project partners for the biochemical research project to develop an immunization method of cattle against methanogen archaea are located in those very countries.

Second, all three countries belong of the ANNEX-I states. This means that, due to their larger economies and higher levels of consumption, they emit a significant amount of GHGs, including CH_4, and therefore bear a greater responsibility with regard to climate change and its mitigation.

Third, although all three are categorized as ANNEX-I states, there are some notable differenze between this states, as e.g. in their size in terms of area, population, and economy (the US. and Germany as leading nations of the world) or in their predominant way to raise animals, particularly cattle (e.g. in New Zealand (NZL) pasteur grazing is dominant in Germany loose housing (**?**, p. 155; **?**, Table 510)).

The time period considered in this bachelor's thesis is determined, on the one hand, by the data series from the published CRT (see section 2.2.2) and, on the other hand, by the evaluation of the mitigation strategy.

For the latter, the decade 2010–2019 is relevant, which is also relevant in Saunois et al. (2025), well as the year 2030 as the target year for the emission targets of the GMP.

2.2.2 CRT Data

Common reporting table (CRT) is the data source for the calculations performed. The CRTs were downloaded from the UNFCCC (**?**).

For Germany and New Zealand, the published data range from 1990 to 2024, and for the United States, from 1990 to 2022. Where is one CRT for one year is published.

For the analysis of methane emissions from enteric fermentation processes involving cattle, the "Table3" and "Table3.A" sheets of the .xlsx files were used. These show only the values for the year it was published for.

The "TableS3" sheet serves as the basis for analyzing methane emissions from the various sectors in each country as well as total emissions. This sheet lists not only the values for the current year but also all past years, so the CRT for 2024 and 2022 for the U.S. also includes all values for the remaining years.

In general, the published figures are always valid for or calculated based on the entire year.

Germany and New Zealand have defined two subcategories for cattle according to Buendia et al. (2019): “dairy cattle” and “non-dairy cattle.” The United States, on the other hand, chose to define its own subcategories for cattle, so that “dairy cattle” and “non-dairy cattle” were further subdivided. Section section 3.2.1 explains how the data was processed to ensure that all three countries are comparable with one another.

The used variables from the CRT table are given in following units for each animal category:

- CH_4 emission in kt CH_4
- Population size in 1000s
- GE in MJ head^{-1} year^{-1}
- Ym in %
- DE in %

2.3 Program

All analyses and processing of the data as well as the visualization took place in R (version 4.6.1). The main package used for processing and analyses was "tidyverse" and "purrr". The visualization was performed via "ggplot". For more information on that, check the R scripts.

3 METHODS

3.1 Biochemestry

3.2 Geography

The general methodology for the processing, analyses and visualization of the CRT data in R was to build a function and then apply these with the interested variables and data from the specific countries.
In the following part, the general procedure is described in the part below. For a more detailed overview, please check the R scripts.

3.2.1 Processing CRT-Data

Cattle enteric fermentation
For Germany and New Zealand all given data for cattle and its subcattegories "dairy cattle" and "non-dairy cattle" were imported from the "Table3.A" sheet of th CRT tables for each year, and processed so, that for each year of each country a table was created with one column with the year, one column, with the cattle categories and the following columns are for the measured or calculated values. Afterwards, all one year tables of one country were merged, so that one country has one table, where the values for the cattle categories from 1990 to the last published year are present.
For the U.S. there are no direct "Dairy cattle" and "Non-dairy cattle" categories, but more detailed one. So after the import of the same value columns for each cattle category as described before, these more detailed subcategories have to be merged in the same 2 as for Germany and New Zealand. For this purpose, the official definition and classifications from the U.S. (**??**, Chapter 5, p. 6) were used. DE, GE and Ym were merged as a weighted mean with the populations as weight. The methane emission were summed.

Methane emission for each sector
For sector-specific emissions, only the “TableS3” sheet was used for the most recent CRT (for the U.S., the year 2022; for Germany and New Zealand, the year 2024). Here, all sectors and their subcategories were imported for all years, as were the total emissions. After processing the data, only the methane emission values for the sectors in general were retained, as well as the total methane emissions for each country.

3.2.2 Calculation of methane emissions from enteric fermentation processes in cattle

The total methane emission for cattle and its subcategories dairy and non-dairy was not directly extracted from the published common reporting table (CRT) for the countries, but calculated from the indicators methane conversionfactor (Ym), gross energy intake (GE) and the population number of the animal categories. Therefore, the CH_4 emissions were calculated for dairy and non-dairy cattle following the eq. (1). The methane emission for cattle than is the sum of dairy and non-dairy.

$$CH_{4,T} = \frac{GE \cdot (\frac{Ym}{100}) \cdot 365}{55.65} \cdot \frac{N_T}{10^6} \tag{1}$$

where:

- $CH_{4,T}$ = yearly methane emission of a animal category [kt CH_4 year^{-1}]
- GE = gross energy intake [MJ head^{-1} day^{-1}]
- Ym = methane conversion factor [%]
- 55.65 = energy content of CH_4 [MJ kg^{-1}]
- N_T = population-size of animal category
- 10^6 = conversion-factor kg to kt

3.2.3 Share of cattle methane emissions from enteric fermentation

The share of enteric fermentation was claculated for its fraction on the total amount of anthropogen produced CH_4 in a country and on the methane emissions resulting from the agriculture sector. For this purpose the published data in the CRT was used for each country.

3.2.4 Calculation of saved methane

To calculate the potential methane savings resulting from the mitigation strategy under investigation (vaccination of cattle), various success scenarios are defined: 10, 20, 30, 40, and 50

These will have an impact on two indicators used in Buendia et al. (2019). Ym, which represents the proportion of energy or feed consumed by cattle that is converted into methane by that microbiome, and DE, assuming that the reduced amount of feed energy converted into methane now benefits the animal during digestion, meaning that the percentage of feed digestibility for the animal increases.

The values of the resulting Ym (Ym_{Red}) and DE (DE_{Red}) are calculated as in eq. (2) and eq. (3).

$$Ym_{Red} = Ym \cdot \left(1 - \frac{Red}{100}\right) \tag{2}$$

where:

- Ym_{Red} = adjusted Ym for the investigated reduction scenario [%]
- Ym = original methane conversion factor from CRT [%]
- Red = investigated reduction scenario [%]

$$DE_{Red} = DE + \left(Ym \cdot \frac{Red}{100}\right) \tag{3}$$

where:

- DE_{Red} = adjusted DE for the investigated reduction scenario [%]
- DE = Digestibility of feed as a fraction of gross energy intake [%]
- Red = investigated reduction scenario [%]

The eq. (1) presents the relation of Ym and methane emission of cattle already. DE, on the other hand, is named in three different equations in the calculation of Buendia et al. (2019), shown in the eq. (4) through (6) with little adjustments. It has his impact on produced methane through his role in calculate the GE of the animal, which has a linear relation towards methane emission (see eq. (1)).

$$GE = \frac{\left(\frac{NE_x}{REM}\right) + \left(\frac{NE_y}{REG}\right)}{\left(\frac{DE}{100}\right)} \tag{4}$$

where:

- GE = gross energy intake [MJ head^{-1} day^{-1}]
- NE_x = sum of the net energy required for maintenance (NE_m), for the activity (NE_a), for lactation (NE_l), for the work (NE_work) and for the pregnancy (NE_p) of the animal [MJ head^{-1} day^{-1}]
- REM = ratio of net energy available in a diet for maintenance (Equation (5))

- NE_y = sum of the net energy required for growth and for wool production [MJ head^{-1} day^{-1}]
- REG = ratio of net energy available for growth in a diet (Equation (6))
- DE = Digestibility of feed as a fraction of gross energy intake [%]

$$REM = \left[1.123 - (4.092 \cdot 10^{-3} \cdot DE) + (1.126 \cdot 10^{-5} \cdot DE^2) - \left(\frac{25.4}{DE}\right)\right] \tag{5}$$

where:

- REM = ratio of net energy available in a diet for maintenance
- DE = Digestibility of feed as a fraction of gross energy intake [%]

$$REM = \left[1.164 - (5.16 \cdot 10^{-3} \cdot DE) + (1.308 \cdot 10^{-5} \cdot DE^2) - \left(\frac{37.4}{DE}\right)\right] \tag{6}$$

where:

- REG = ratio of net energy available for growth in a diet
- DE = Digestibility of feed as a fraction of gross energy intake [%]

To calculate the new methane emission for cattle corresponding to the reduction scenario now, GE has to be recalculated with the adjusted DE. Afterwards eq. (1) can be used again to calculate the adjusted CH_4 emission with the adjusted GE (GE_{Red}) and Ym_{Red}.
For calculation the GE_{Red} DE_{Red} can not just simply be put in the eq. (4), because of the values for NE_x and NE_y. Although the values for the individual net energy intake categories (see eq. (4)) should not change, these are not given in the published CRTs of the countries. To solve this problem, I assume two things:

1. The net energy required for wool production an irrelevant categorie is for the investigation of cattle.
2. Most of the animal in the population of dairy and non-dairy cattle are fully grown or close to it. Here, too, the IPCC only classifies calves "dairy" or "non-dairy" once they started eating solid food. Because of that the net energy needed for growth will overlooked and set to 0.

Now, we can slim down the eq. (4) and rearrange it to calculate a fixed value for NE_x for a specific year, because the part with NE_y turns zero (see eq. (7)).

$$NE_x = GE \cdot REM \cdot \left(\frac{DE}{100}\right) \tag{7}$$

$$GE = \frac{\left(\frac{NE_x}{REM}\right)}{\left(\frac{DE}{100}\right)} \tag{8}$$

where:

- NE_x = sum of the net energy required for maintenance (NE_m), for the activity (NE_a), for lactation (NE_l), for the work (NE_work) and for the pregnancy (NE_p) of the animal [MJ head^{-1} day^{-1}]
- GE = gross energy intake [MJ head^{-1} day^{-1}] (eq. (4))
- REM = ratio of net energy available in a diet for maintenance (Equation (5))
- DE = Digestibility of feed as a fraction of gross energy intake [%]

Now, GE_{Red} can be calculated for a given year using the fixed values for NE_x and DE_{Red}, when NE_y is assumed to be 0 (see eq. (8)). The calculated gross energy intake is then substituted into eq. (1) along with the corresponding Ym_{Red} value. This yields the methane emissions ($CH_{4,R}$) for a cattle category under a reduction scenario for a specific year.
The saved methane emissions ($CH_{4,S}$) for a specific reduction scenario are calculated by subtracting the original value for methane emissions from the value adjusted for the scenario (see eq. (9))

$$CH_{4,S} = CH_{4,T} - CH_{4,R} \tag{9}$$

where:

- $CH_{4,S}$ = saved CH_4 through reduction scenario [kt CH_4 year^{-1}]
- $CH_{4,T}$ = yearly methane emission of a animal category [kt CH_4 year^{-1}]
- $CH_{4,R}$ = yearly methane emission of a animal category for a corresponding reduction scenario [kt CH_4 year^{-1}]

3.2.5 Prediction of CH_4 emission until 2030

Two steps are required to calculate total methane emissions, as well as methane emissions for three reduction scenarios (10, 30, and 50) through the year 2030.

First, total methane emissions without emissions from EnterFe processes in cattle are calculated and then extrapolated.

Second, to calculate the methane emission from enteric fermentation (EnterFe) in cattle, the values for DE, Ym, Population size and NE_x were predicted for each subcategorie ("dairy" and "non-dairy cattle") until 2030. For that, NE_x had to be first calculated (see eq. (7)). From the predicted values for these indicators the cattle EnterFe CH_4 emission could be calculated (via eq. (8) and (1)).
Also the the methane emission values for three different reduction scenarios (10, 30 and 50) were calculated with these values using the eq. (2), (3), (8) and (1).
Afterwards, the emissions of the subcategories were summed to the CH_4 emission for cattle resulting from enteric fermentation processes.

Third, the predicted emission values for total without cattle enteric fermentation were added with the calculated emission values for cattle enteric fermentation to achieve the total methane emission of a country.

The forecast was made using a linear regression, with the value to be predicted depending on the respective year. However, the regression was not based on the entire time period, but rather on three different time intervals (the last 5, 10, and 15 years). For each linear model, both the confidence intervall (CI) with a confidence level of 95% and the total emissions were calculated. In addition, the R^2 value was determined for each model.

The extrapolation extends from 2025 to 2030 for New Zealand and Germany, and from 2023 to 2030 for USA.

4 RESULTS

4.1 Biochemistry

All following experiments were done with *Methanobrevibacter boviskoreani AbM4* (*M. boviskoreani*) and *Methanobrevibacter thaueri CW^T* (*M. thaueri*), in some *Methanobrevibacter wolinii SH^T* (*M. wolinii*) and *Methanobrevibacter millerae ZA-10^T* (*M. millerae*) were investigated as well.

4.1.1 Cultivation

Media

For cultivation two liquid, complex media were tested, the DSM 1523 and the DSM 119. Both media are suitable for the cultivation of *M. thaueri* and *M. boviskoreani*, although some specific behavior could be detected.
The *M. thaueri* grew more slowly and not as dense as the *M. boviskoreani*. Also the *M. thaueri* started to lyse rapidly in a few days after full growth. This was observed much more prominently in DSM 1523 than in 119.
M. boviskoreani on the other hand did not show the described lysing behavior but precipitated in both media. These precipitates were not completely dissolvable through shaking. *Here I also could implement the Microscopy picture of the precipitate?*

Substrate

Besides the general cultivation with a H_2/CO_2 headspace, it was tested if *M. thaueri* and *M. boviskoreani* could use ethanol (EtOH) as an alternative source of reducing potential for growth, as a liquid option to hydrogen. For the tested concentration of EtOH (10 mM, 20 mM, 50 mM, 100 mM, 200 mM) no visual growth could be detected. In einer literatur steht für typstamm bovis er wachse auf EtOh, und für Format auch -> das könnte man reintheoretisch noch testen

4.1.2 Cell count

Could also show a microscopy picture of cell counting? -> hänge hier vmtl am besten noch eins in den Anhang, oder soll ich für alle 4 dann ein Bild in den anhang hängen?
Cell counting was done with a phase-contrast microscope and a Neubauer improved counting chamber, only the *M. wolinii* with an optical density at 600 nm (OD600) of 0.387 was counted

with a Buerker counting chamber. In the Buerker, squares were counted, in the Neubauer improved, the group squares were counted, both have a volume of 4 · 10^{-6} ml, as can be seen in LO - Laboroptik GmbH (2026). der part hiervor vmtl eher in die methoden

The cell-number per ml for the cultures was then determined using the following calculation:

$$\textit{cellnumber} \cdot ml^{-1} = \frac{\textit{ø cellnumber} \cdot \textit{dilution factor}}{\textit{Volume of the counted Square}} \tag{10}$$

Table B1 through B6 in the appendix contain the original counted cellnumbers per filled chamber and square. To calculate the cell count per ml, the arithmetic mean was determined for each culture (tab. B7) and substituted into the equation 10 along with the corresponding dilution factor (see ibid.). The results of the calculation were recorded in a table 4.1. The calculation was performed as an example using *M. wolinii* (OD600 = 0.387):

$$\textit{cellnumber} \cdot ml^{-1} = \frac{216.375 \textit{ cells} \cdot 20}{4 \cdot 10^{-6}\ ml} = 1,081,875,000 = 10.819 \cdot 10^{8}$$

Table 4.1: *Cellnumber per ml of different Methanobrevibacter (M.) culture for corresponding optical density at 600 nm (OD600).* For counting the cells a phase-contrast microscope was used and a Buerker or Neubauer improved counting chamber at a 400x magnification. The cell count per ml was calculated using the equation 10, and the values were taken from the table B7.

culture	OD600	cellnumber · ml^{-1}
M. wolinii	0.387	10.819 · 10^{8}
M. wolinii	0.396	10.672 · 10^{8}
M. thaueri	0.235	5.603 · 10^{8}
M. thaueri	0.116	3.644 · 10^{8}
M. millerae	0.378	8.880 · 10^{8}
M. boviskoreani	0.287	9.272 · 10^{8}

A clear trend can be observed for the cellnumber per ml of the different *Methanobrevibacter* (*M.*) culture throughout the table 4.1: higher OD600 values correspond to a higher cell count per ml. At the lowest measured OD600 of 0.116 (*M. thaueri*), 3.644 · 10^{8} cells per ml was determined, and at the highest OD600 of 0.396 (*M. wolinii*), 10.672 · 10^{8} cells per mll was determined.

Nevertheless, differences between the organisms are also apparent. For example, *M. boviskoreani* (OD600 = 0.287) has a higher cell count per mL (9.272 · 10^{8}) than *M. millerae* (OD600 = 0.378 and celnumber per ml = 8.880 · 10^{8}) despite having a lower OD600 of nearly 0.1. Furthermore, both *M. wolinii* cultures and *M. millerae* have similar OD600 values between 0.378 and 0.396, but those of *M. wolinii* are around 2 · 10^{8} cells per ml higher (10.819 or 10.672 · 10^{8} to 8.880 · 10^{8} cells per ml).

The *M. wolinii* cultures are quite close to each other in terms of cell count per ml, with the higher value being determined for the culture with the lower OD600.

Besides *M. wolinii*, two cultures of *M. thaueri* were also examined to determine cell count. Here, one culture has an OD600 of 0.235 that is slightly more than twice as high as the other one with 0.116, but a cell count per ml that is slightly less than twice as high ($5.603 \cdot 10^8$ to $3.644 \cdot 10^8$ cells per ml).

4.1.3 Plating

Spread plating

100 µl (nochmal im Laborbuch nachschauen) of a full-grown culture were spread plated as duplicates on a DSM 1523 solid media (1.5% agar) under anaerobic condition in a glovebox. In figure 4.1 the plating results are shown.

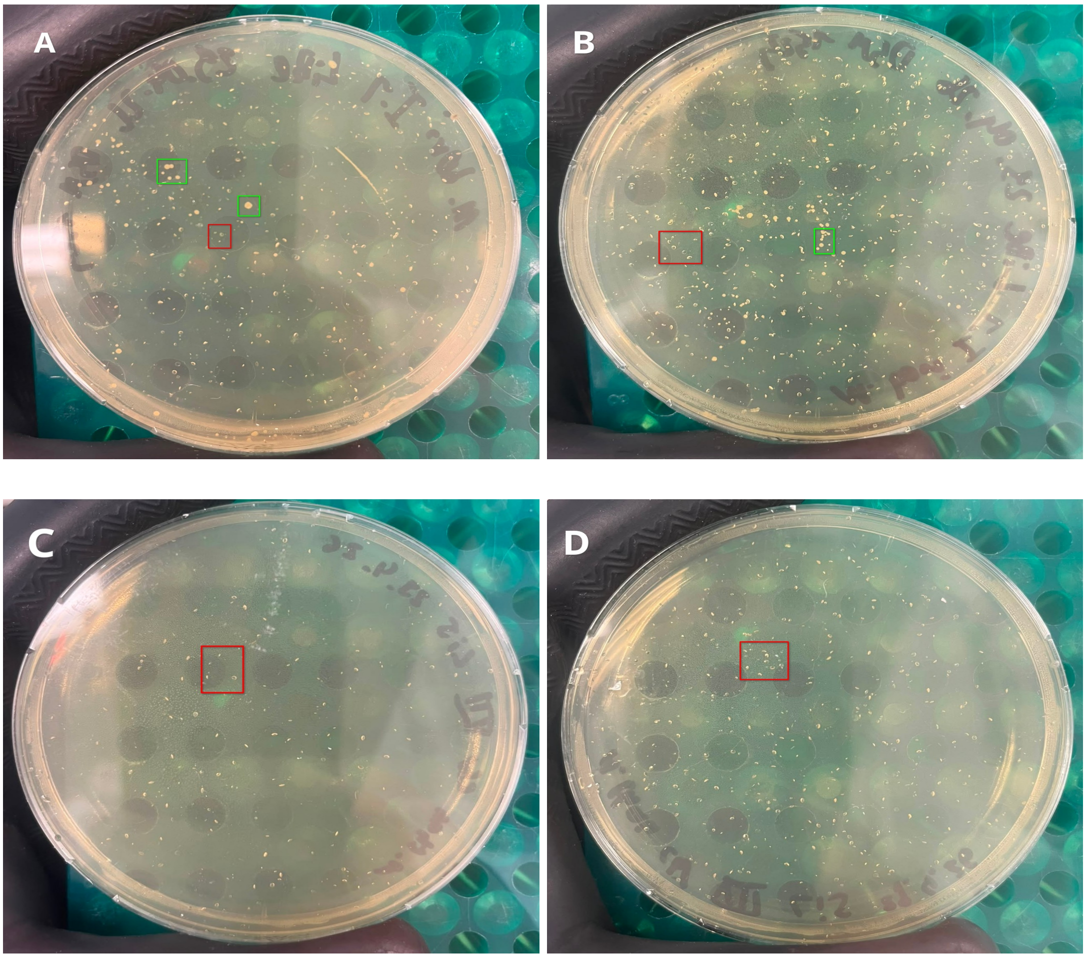

Figure 4.1: Spread plating results of *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani AbM4* (*M. boviskoreani*). 100 µl of a full-grown culture was plated on DSM 1523 (1.5% agar) in duplicates. Red rectangles highlight gas bubble spots, green rectangles grown colonies. A: *M. boviskoreani*, B: *M. boviskoreani*, C: *M. thaueri*, D: *M. thaueri*

Various spots can be seen on all 4 plates. Most of these are gas bubbles, as indicated by the smaller size and transparency of the spots (see red rectangles in Figure 4.1). In contrast, a few spots are opaque, larger, round, slightly milky, and yellow (see green squares ibid.). These are grown colonies. Das die einen Kolonien sind und die anderen Gasblasen ist streng genommen ja bereits eine Interpretation, finde es aber in dem Kontext recht nervig und wenig förderlich für die Verständlichkeit, das nur vage zu bezeichnen als milchig und weniger milchig. Wie siehst du das Max?

No mature colonies of *M. thaueri* were detected on the two plates. *M. boviskoreani*, on the other hand, grew on both plates. Individual colonies were picked from the plates, propagated in liquid DSM 1523, and verified as *M. boviskoreani* via 16S-Sequencing.

Pour plating

The pour plating was not successful for 100 µl of a full-grown *M. thaueri* culture with DSM 119 (1% agar?) as you can see in fig. A1. This experiment was also performed as duplicate.

Roll Plating

200 µl of full-grown *M. thaueri*, *M. wolinii* and *M. boviskoreani* culture have been roll plated in DSM 1523(1% agar?).

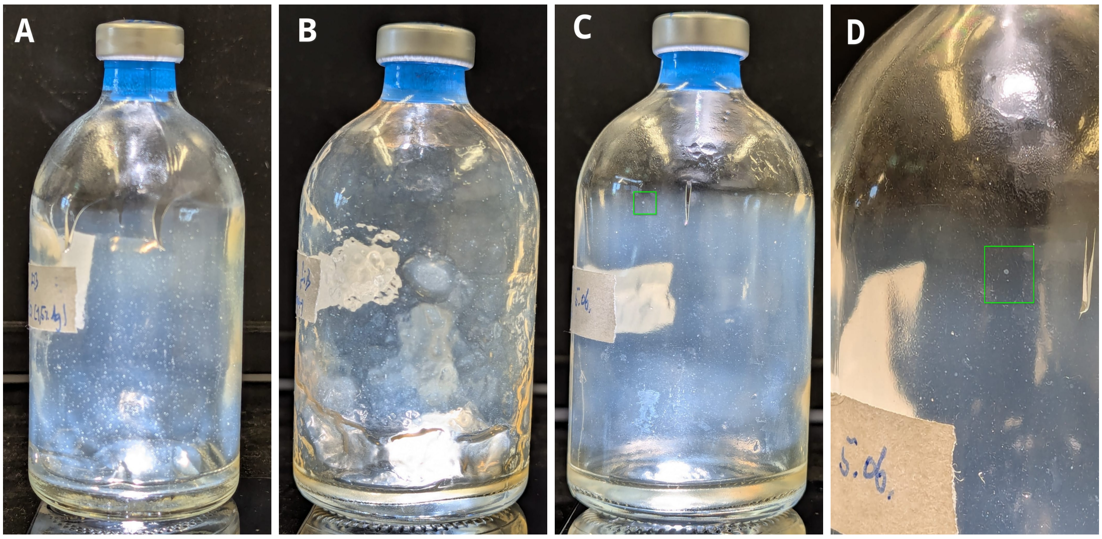

Figure 4.2: Roll plating of *Methanobrevibacter thaueri CW*^{T} (*M. thaueri*), *Methanobrevibacter wolinii SH*^{T} (*M. wolinii*) and *Methanobrevibacter boviskoreani AbM4* (*M. boviskoreani*). 200 µl of a full-grown culture were used. The green rectangle marks the same colony in C and D. A: *M. boviskoreani*, B: *M. wolinii*, C: *M. thaueri*, D: *M. thaueri* zoomed in on colony

In all three bottles CFU were formed (see fig. 4.2). *M. thaueri* showed notable less colonies (see table 4.2: 15 for 2432 or 3536), but they were larger than the little dots for the other cultures(fig. 4.2). Another difference between the bottle of *M. wolinii* (fig. 4.2 B) and the bottles of *M. thaueri* (ibid. C, D) and *M. boviskoreani* (ibid. A) is the agar structure. At *M. wolinii* it is not as smooth and a bit more clumpy. In diskussion auf unterschiedliche technick beim rollen verweisen -> schlussfolgern wie am besten zu performen ist

However, besides the growing spots inside the bottles, shown in figure 4.2, also the H_2/CO_2 headspace changed. So approximately every two days the gas pressure in tbottle dropped from about 1.1 bar to below 0.8 bar and hat to be refilled. weist auf anaerobic life hin, ob die spots jetzt wirklich

Verification via 16S-Sequencing, as is done with spread plating, is still pending.

Even though the OD600 of the cultures used was not measured prior to plating, the plating efficiency was determined. This was done on the assumption that the cultures that appeared visually full-grown had an OD600 approximately equal to those listed in tab. 4.1, with the higher value for cell count per mL always being used.

$$\textit{Plating efficiency [ppm]} = \frac{\textit{counted CFU}}{\textit{number of used cells}} \cdot 10^6 \tag{11}$$

The calculation was based on equation 11 and was performed as follows, for example:

$$\textit{Plating efficiency} = \frac{2432}{0.2\ ml \cdot 10.819 \cdot 10^8\ cells \cdot ml^{-1}} \cdot 10^6 = 11.24\ ppm$$

Table 4.2: Counted colony forming unit (CFU) of different *Methanobrevibacter* (*M.*) culture for roll plating technic. 200 µl of a full-grown culture was used for plating. For *M. wolinii* and *M. boviskoreani* was counted a sixteenth of the bottle. Always the higher number of cells per ml from table 4.1 were used as a reference for the cell count in the initial solution.

culture	counted CFU	plating efficiency [ppm]
M. wolinii	2432	11.24
M. thaueri	15	0.13
M. boviskoreani	3536	19.07

4.2 lists the CFU counts per bottle as well as the corresponding plating efficiency. For *M. wolinii* and *M. boviskoreani*, the efficiency is in the lower two-digit ppm range. The efficiency for *M. thaueri* is notable lower, at 0.13 ppm.

4.1.4 Antibiotic sensitivity

Puromycin

M. thaueri and *M. boviskoreani* were cultivated in DSM 1523 media with various concentration of puromycin (Puro): 0 µg/ml, 1 µg/ml, 2 µg/ml, 5 µg/ml, 10 µg/ml and 20 µg/ml. The experiment was performed in triplicates, with each antibiotic (AB) concentration series inoculated with one full-grown pre-culture. The OD600 was taken once a day and plotted logarithmically against time in figures 4.3 and 4.4. For plotting negative measured OD600 values were set to 0.001. The original data and the adjusted OD600 values are in the appendix (see table **??** and **??**). Max die tabellen sind halt extrem lang und gehen dann über drei oder vier seiten, sollen die dann trzdm einfach so in den Anhang? oder kann ich die bei github mit hochladen als csv

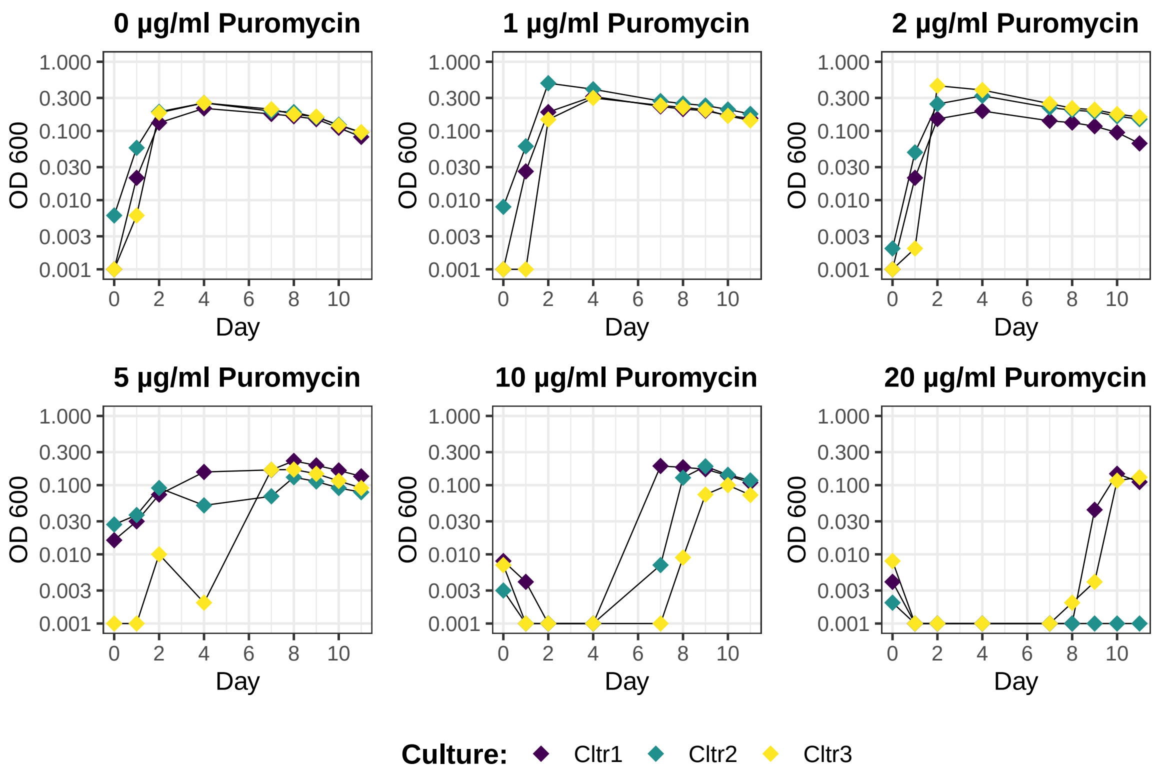

Figure 4.3: Growthcurve in DSM 1523 for *Methanobrevibacter boviskoreani AbM4* (*M. boviskoreani*) for various concentration of puromycin (Puro): 0 µg/ml, 1 µg/ml, 2 µg/ml, 5 µg/ml, 10 µg/ml and 20 µg/ml. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) of the pre-culture were 0.110 for Cltr 1, 0.091 for Cltr 2 and 0.034 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

Fig. 4.3 shows that *M. boviskoreani* exhibits a very similar growth pattern for Puro concentrations of 0, 1, and 2 µg/ml. Exponential growth occurs within the first two days. This is followed by a delayed growth phase, such that the growth maximum is reached between days two and four at an OD600 of around 0.3. Until the final measurement day (day 10), the OD decreases continuously to an OD600 value of around 0.1. One difference that can still be noted here, however, is that the variation between the triplicates at 0 µg/ml is smaller than that observed with Puro. In the latter case, culture 3 does not begin exponential growth until one day later as the other triplicates.

At 5 µg/ml, the growth curves for Cltr. 1 and Cltr. 2 are generally similar to those just described. However, the two cultures do not grow as quickly. Cltr. 3 shows delayed growth and exhibits greater fluctuations in the OD600 values. Thus, all three triplicates here reach their maximum growth only after 8 days. Nevertheless, the maximum OD600 values here are also slightly lower than in the previous growth curves. Thereafter, the OD600 values decrease continuously.

At 10 and 20 µg/ml Puro concentrations, exponential growth begins with a notable delay. At 5 µg/ml, this occurs between 4 and 7 days. The maximum is reached between 7 and 10 days, with maximum OD600 values ranging from approximately 0.1 to 0.25. At 20 µg/ml, one replicate shows no growth at all by the end of the measurement (Cltr. 2). The other two replicates do not begin exponential growth until day 8 or 9. The maximum is reached on day 10 at approximately 0.1. By day 11, a slight decrease in the OD600 values can already be observed.

The growth curves of *M. thaueri* for Puro concentrations of 0 and 1 µg/ml reach their OD600 maximum after 3 days, similar to *M. boviskoreani* (see Figure 4.4). However, the maximum does not exceed 0.3 and is closer to 0.2 than to 0.3.
After that, the OD600 decreases rapidly again, so that at 0 µg/ml, Clt. 1 and 2 already have an OD below 0.003 after 9 days, and Cltr. 3 is also just barely above 0.01 after 10 days. For 1 µg/ml, the final measured OD values range between 0.001 and 0.05.

For 2 µg/ml puromycin, Figure 4.4 shows growth for all three triplicates, reaching its maximum on day 3. However, the increase in OD600 values is significantly lower. Thus, the maximum for Cltr. 2 is below 0.03, and for the others it is around 0.1. This is followed by another decrease in the values, so that the final values range between 0.001 and 0.03.

For the other measured Puro concentrations, in contrast to *M. boviskoreani*, no growth was observed with one exception (Cltr. 3 at 5 µg/ml) (see Fig. 4.3 and 4.4), with the OD600 values remaining below 0.003. For Cltr. 3 at a Puro concentration of 5 µg/ml, an OD600 of more than 0.1 was measured on day 3, but this value dropped back to 0.003 by day 6.
In der Diskussion drauf verweisen, dass man aus der Kurve für thau auch die Lyse gut erkennt, die auch bei den Beobachtungen bei der Kultivierung mit benannt wurden.

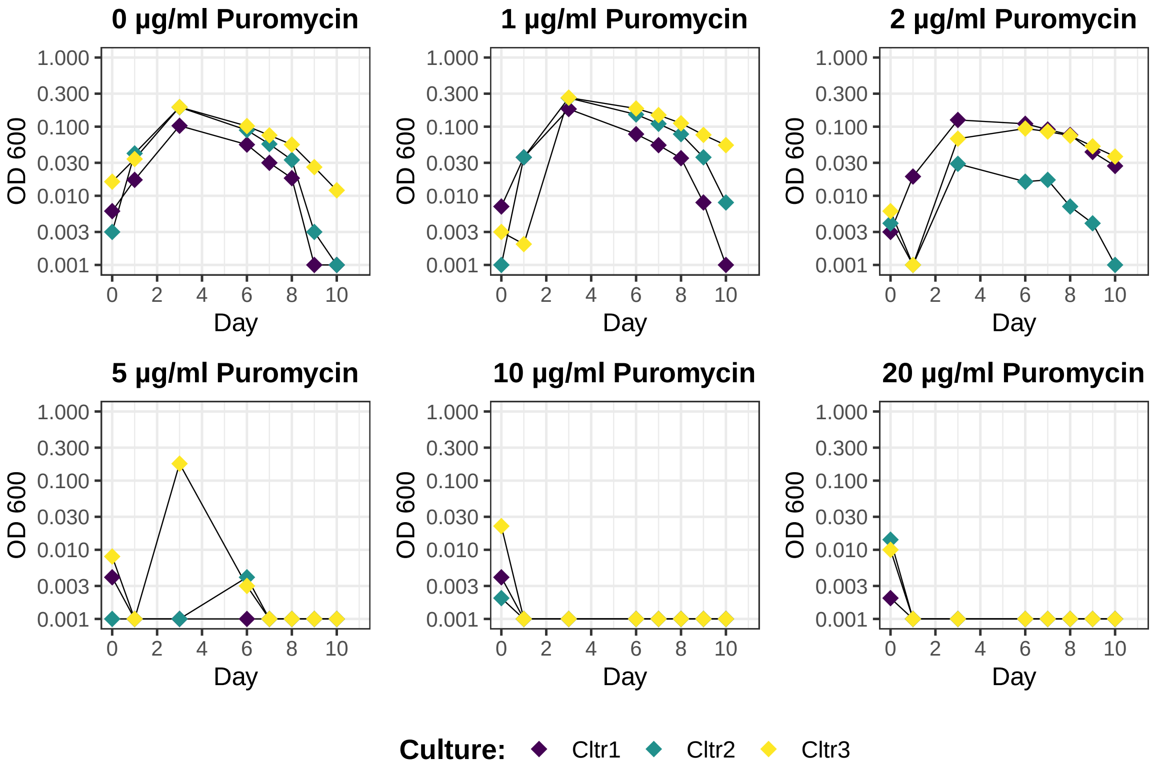

Figure 4.4: Growthcurve in DSM 1523 for *Methanobrevibacter thaueri CW^T* (*M. thaueri*) for various concentration of puromycin (Puro): 0 µg/ml, 1 µg/ml, 2 µg/ml, 5 µg/ml, 10 µg/ml and 20 µg/ml. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) of the pre-culture were 0.154 for Cltr 1, 0.173 for Cltr 2 and 0.120 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

Cycloserin

In duplicates it was tested if 200 µg/ml cycloserin (Cyc-Se.) would inhibit the growth of *M. thaueri* in DSM 1523. In all four bottles(two negativ and two with 200 µg/ml Cyc-Se.) visual growth could be detected

4.1.5 Nucleotide-baseanalogica sensitivity test

8-Azahypoxanthin

Similar to the AB sensitivity test, a growth experiment was conducted for the base analogs 8-Azahypoxanthin (8-Aza) for *M. thaueri*, *M. boviskoreani*, *M. millerae*, and 6-Azauracil (6-Aza) for *M. wolinii*. One difference is that the base analogs do not dissolve in H_2O, so they were dissolved in dimethylsulfoxid (DMSO). Furthermore, different concentrations were selected: 0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml, and one with 1 ml DMSO.

The OD600 was measured and plotted on a logarithmic scale against time (see Figures 4.5 through 4.8). The original measured data is in the appendix (Fig. X).

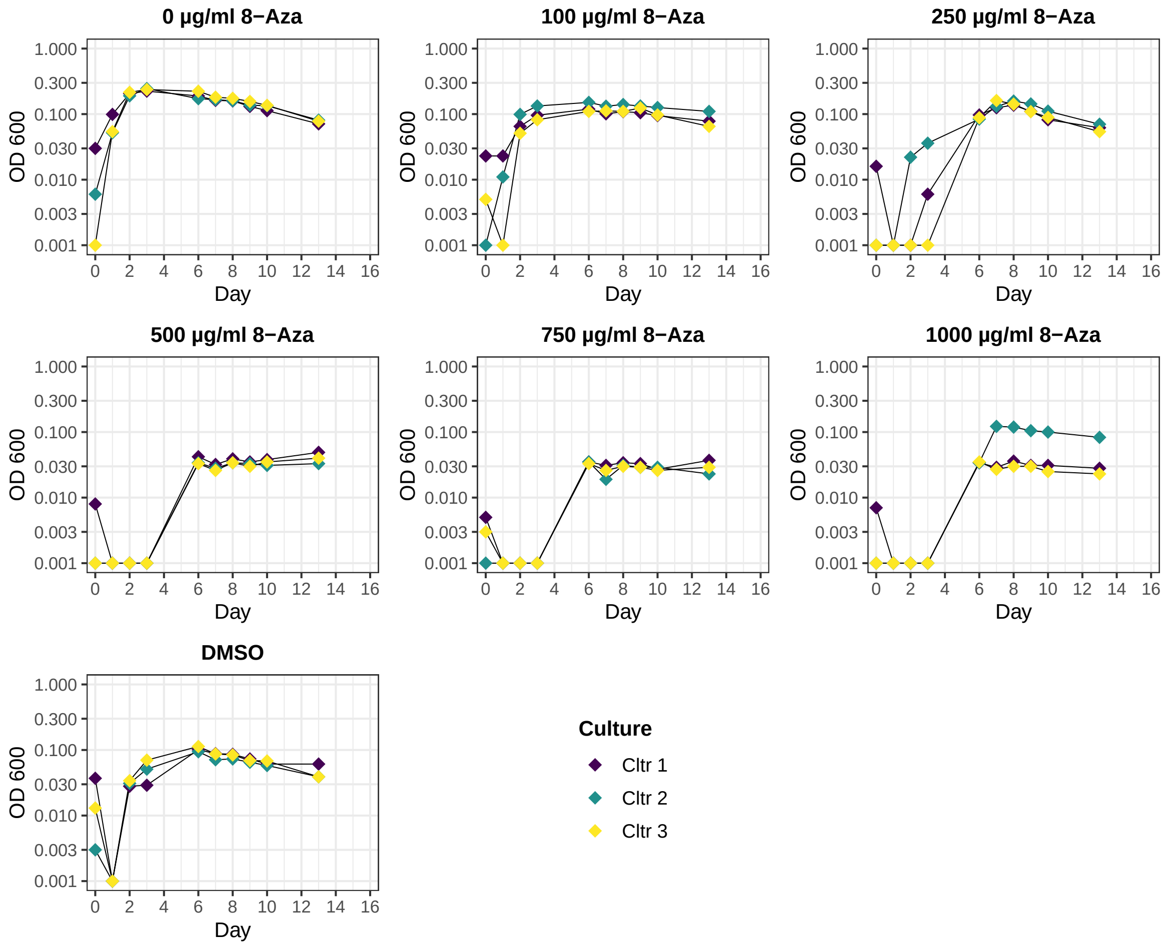

Figure 4.5: Growthcurve in DSM 1523 for *Methanobrevibacter boviskoreani AbM4* (*M. boviskoreani*) for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 8-Aza. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) of the pre-culture were 0.096 for Cltr 1, 0.105 for Cltr 2 and 0.152 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values lass than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

4.5 shows a growth curve for the positive control (0 µg/ml 8-Aza) of *M. boviskoreani* that is similar to that of the positive control in Fig. 4.3. Thus, exponential growth occurs within the first two days. On days 3 through 4, the OD600 reaches a maximum of around 0.3, followed by a continuous decline with a final OD600 of just under 0.1 on day 13.

The solvent control (DSMO), in which no base analogs were added but only 1 ml of the solvent DMSO, showed a different growth curve. Initially, none of the triplicates begin to grow immediately; instead, the OD600 values first decrease. On day 3, slight growth begins, reaching a maximum of 0.1 on days 4–6. Afterward, the values decline to approximately 0.03 on day 13.

At the lower concentrations of 8-Aza (100 and 250 µg/ml), *M. boviskoreani* in fig. 4.5 a growth curve similar to that of the corresponding positive control. However, at 100 µg/ml 8-Aza, the maximum OD600 value is reached within 1 to 3 days, but it is approximately 0.1 instead of just under 0.3. 0.1 is also the maximum OD600 value for the 250 µg/ml concentration; here, growth does not begin until 2 to 4 days later, so that the maximum is not reached until 7 days later.

The three remaining 8-Aza concentration solutions (500, 750, and 1000 µg/ml) exhibit a very similar growth curve. An OD600 below 0.01 was measured for *M. boviskoreani* during the first 4 to 6 days. Then there was an increase to an OD600 of 0.03, which remained constant until the final measurement day (day 13). An exception is Culture 2 in the 1000 µg/ml 8-Aza culture. This culture grows from Day 3 to Day 5 to an OD600 of 0.1, which then decreases slightly until the end of the measurement series.

Figure 4.6 visualizes the growth curves for *M. thaueri*. The first thing that stands out in comparison to the other archaea (fig. 4.5, 4.7, and 4.8) is that the baseline OD600 is higher, at approximately 0.03. With the exception of the positive control (0 µg/ml 8-Aza), no significant increase in the OD600 values was observed. Instead, there was a fairly constant fluctuation around the initial value of 0.03 until the last measurement day (Day 16). An exception here is the triplicate Cltr. 3 in the DMSO sample. Here, an OD600 of 0.3 was measured on Day 2, but it fell back to 0.03 by the next measurement day (Day 4). At the end of the measurement series, this triplicate also ended at approximately 0.07 instead of 0.03 like the other triplicates.

In the positive control, slight growth of *M. thaueri* was detected. However, the growth curve differs quite noticeably from that of the positive control for *M. thaueri* in fig. 4.4. In addition, there is considerable heterogeneity among the individual triplicates.
For example, Cltr. 1 begins to grow slightly from the first day and reaches its maximum OD600 between 0.1 and 0.3 on day 9 (see fig. 4.6). By the end of the measurement, the OD600 decreases continuously to 0.1.
Cltr. 3 begins to grow on Day 8, reaching a maximum OD600 of approximately 0.2 on Day 12. Between Days 13 and 14, the last triplicate then begins to show slight growth, reaching

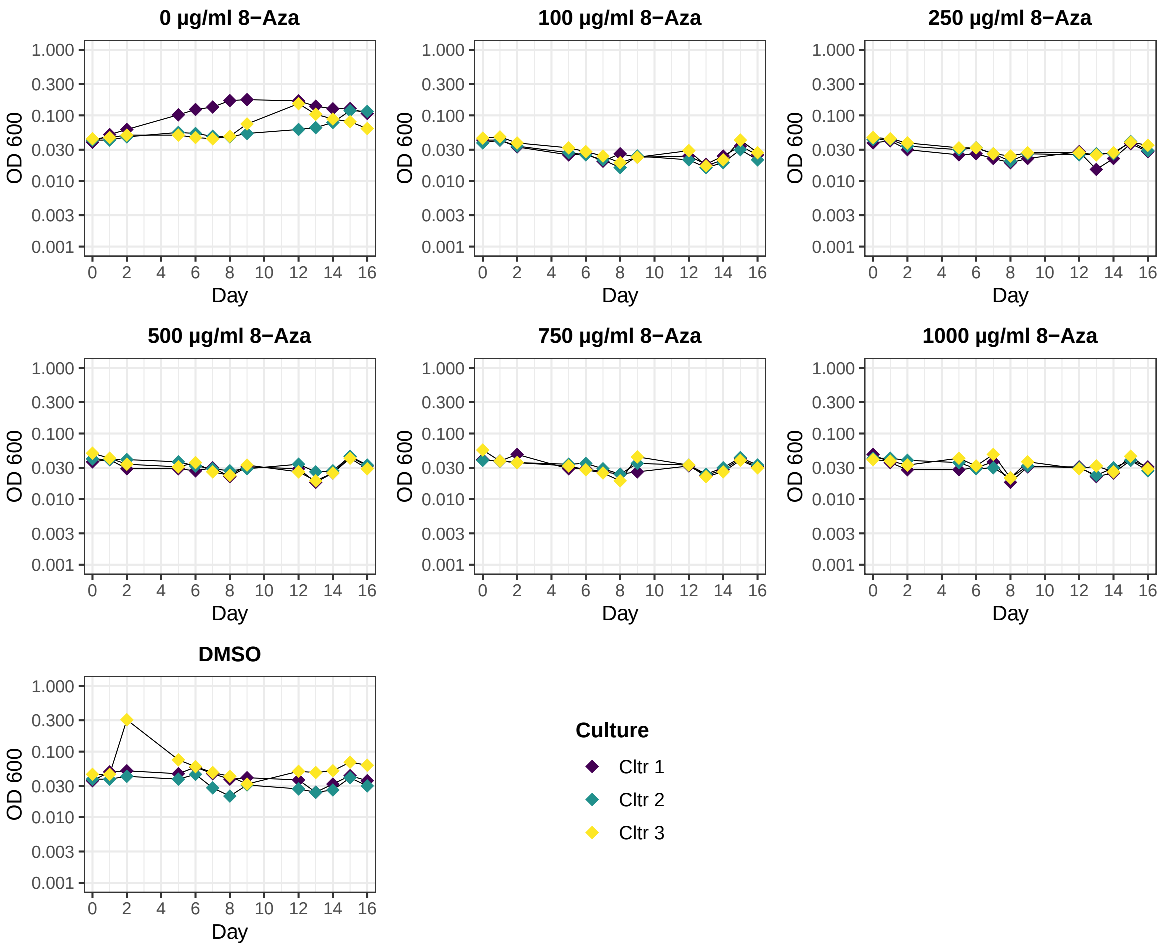

Figure 4.6: Growthcurve in DSM 1523 for *Methanobrevibacter thaueri CW*^{T} (*M. thaueri*) for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 8-Aza. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) of the pre-culture were 0.125 for Cltr 1, 0.148 for Cltr 2 and 0.210 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values lass than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

an OD600 of just over 0.1 on Day 16.

The growth curves of *M. millerae* for the positive control (0 µg/ml 8-Aza) and the solvent control (DMSO) have a similar shape for all triplicates, with the measured OD600 being reached on day three, followed by a continuous decline until the end of the measurement series (day 14). However, the curves differ in their OD600 values. The positive control reaches a maximum value of 0.3, while the solvent control reaches just under 0.1. This also results in a significantly lower final OD600 measurement for DMSO (below 0.03) and for the positive control (just under 0.1).

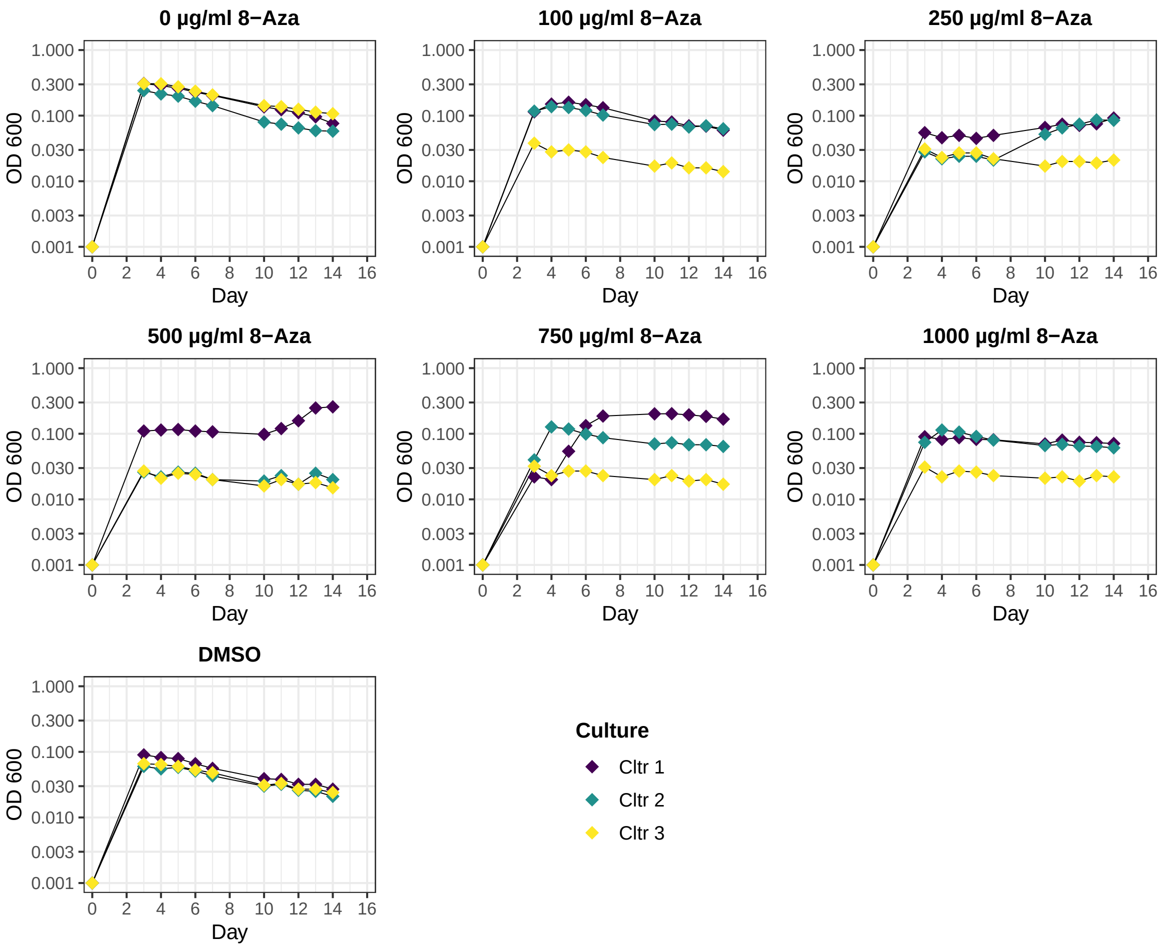

Figure 4.7: Growthcurve in DSM 1523 for *Methanobrevibacter millerae ZA-10*^{T} (*M. millerae*) for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 8-Aza. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) of the pre-culture were 0.366 for Cltr 1, 0.339 for Cltr 2 and 0.281 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values lass than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

Clear trends for the other samples are generally difficult to discern at first glance; however, most triplicates do not reach an OD600 higher than 0.1.

The 100 µg/ml 8-Aza generally follows the curve shape of the positive and solvent controls, although triplicates Cltr. 1 and 2 reach their maximum OD600 value on Day 4 with a value above 0.1, while Cltr. 3 does not exceed 0.03.
For the 250 µg/ml concentration, the triplicates have a constant OD600 of approximately 0.03, with Cltr. 1 and 2 showing slight growth starting on day 7, reaching approximately 0.1 by the end of the measurement. The triplicates for Ctr. 2 and 3 at the 500 µg/ml concentration show a constant OD600 value of 0.03 starting on Day 3. For Ctr. 1, an OD600 of 0.1 was measured

between Days 2 and 10, which increases to just under 0.3 by Day 13.

For the 750 µg/ml and 1000 µg/ml 8-Aza solutions, replicate Ctrl. 3 exhibits a constant value of 0.03 starting on day 2. At 1000 µg/ml, the other two triplicates show a constant OD600 of just under 0.1 starting on day 2, while at 750 µg/ml, only Cltr. 2 shows this value starting on day 2. The Cltr. 1 triplicate in the 750 µg/ml solution shows a value of 0.03 on days 2 and 3, then increases to an OD600 between 0.2 and 0.3 by day 7. This value decreases slightly by the end of the measurement series.

6-Azauracil

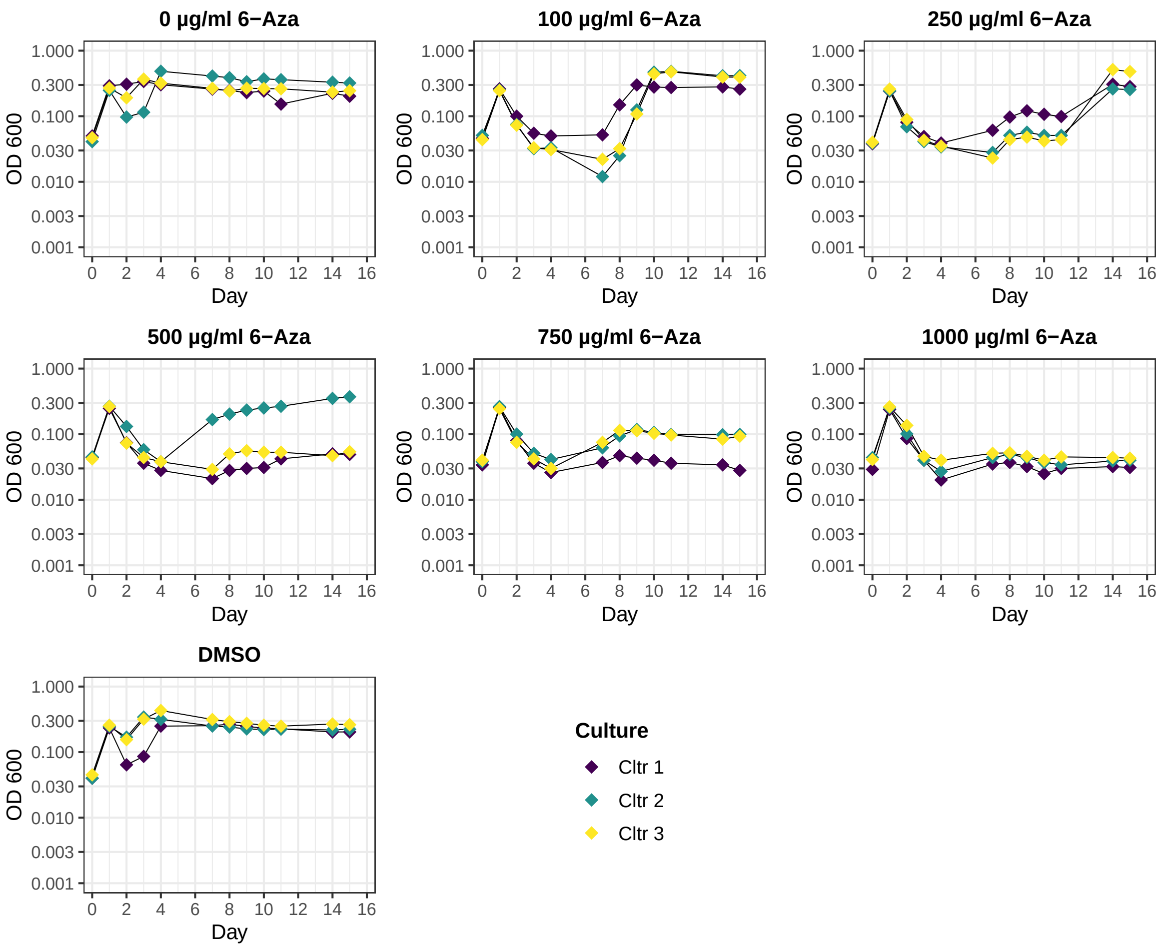

Figure 4.8: Growthcurve in DSM 1523 for *Methanobrevibacter wolinii SH^T* (*M. wolinii*) for various concentration of 6-Azauracil (6-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 6-Aza. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values lass than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

All growth curves for the 6-Azauracil (6-Aza) test series by *M. wolinii* (see fig. 4.8) show a jump in the OD600 values within the first day from an initial value of approximately 0.03 to 0.3, followed by a drop that varies in magnitude depending on the sample and results in different curve profiles.

The triplicate Cltr. 1 of the positive control (0 µg/ml) shows no change after Day 1 and maintains a constant OD600 value of around 0.3 until the end of the measurement series. Cltr. 3 dips slightly to approximately 0.2 on Day 2, then remains constant at 0.3 starting on Day 3. The triplicate Cltr. 2 shows the greatest change after Day 1, remaining at a value of 0.1 on measurement days 2 and 3, before rising to an OD600 of approximately 0.4 on Day 4. The final measured value for this triplicate on Day 15 is approximately 0.3.

The triplicates in the solvent control (DMSO in fig. 4.8) show a more pronounced drop in OD600 values than the positive control. For example, Cltr. exhibits an OD600 of slightly more than 0.1 on Day 2, while Cltr 1 drops to a value below 0.1 on Days 2 and 3. All three triplicates have a fairly stable OD600 of approximately 0.3 starting on Day 4.

In all 6-Aza preparations, a decrease in OD600 values was observed down to a value below approximately 0.03. All triplicates in the 100 µg/ml preparation grew to an OD600 of 0.3 between measurement days 7 and 10, which remained constant until the end of the measurement series.
In the 250 µg/ml preparation, significant growth to an OD600 of more than 0.3 was not observed until between days 11 and 14, even though all triplicates showed a slight increase between days 7 and 8 of approximately 0.03, so that Cltr. 1 had a value of 0.1 and the other two triplicates had values of approximately 0.07.
No further growth was observed for any of the triplicates of the 1000 µg/ml concentration by the end of the measurement series. The same was true for triplicates Cltr. 1 and 3 of the 500 µg/ml 6-Aza concentration and Cltr. 1 at 750 mg/ml.
The OD600 of Cltr 2 in the 500 µg/ml preparation rises from 0.03 on measurement day 4 to approximately 0.2 on measurement day 7, and subsequently to a final OD600 of more than 0.3 by the end of the measurement series.

DMSO

In general, all preparation with DMSO showed the tendency to be more vulnerable to turn red when taking a sample for photometric measurement. In addition, The decolorization also took longer than in the positive control or other experimental conditions without DMSO, sometimes over an hour.

This interaction of DMSO with the DSM 1523 media was investigated more closely in the *M. thaueri* measurement series. Here, one triplicate with DSM 1523 was prepared with just 1 ml DMSO, but without inoculation (DMSO –, see fig. 4.9).

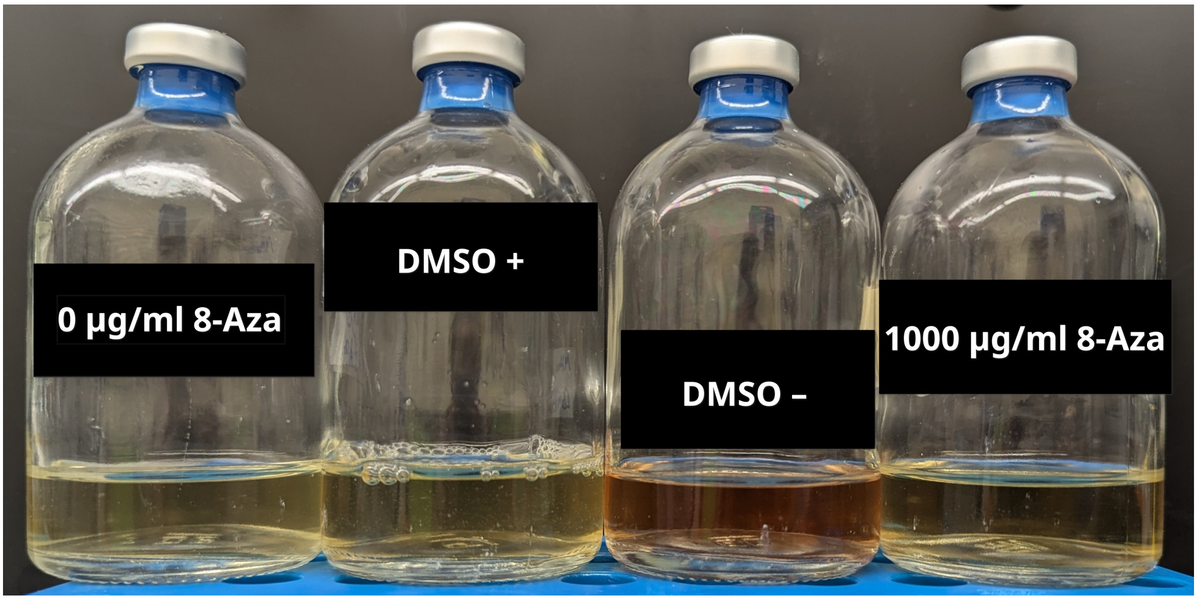

Figure 4.9: Color differences with and without DMSO after 6 days of incubation. Image of various preparations for *M. thaueri* from the same triplicate (Cltr. 3). The positive control (0 µg/ml 8-Azahypoxanthin (8-Aza)) contains only the organism (*M. thaueri*) in DSM 1523, 1000 µg/ml 8-Aza *M. thaueri* and the named base analog, DMSO + *M. thaueri* and 1 ml DMSO; DMSO – also contains 1 ml DMSO but was not inoculated.

After the first few days of incubation, a difference in color was observable among the cultures. Specifically, the DMSO bottles exhibited a red color, in contrast to the yellow color of the other samples in the 8-Azahypoxanthin (8-Aza) measurement series for *M. thaueri*. This is illustrated in fig. 4.9 using selected samples from the triplicate Cltr. 3 after 6 days of incubation. The OD600 was also measured for DMSO, but showed no difference from the baseline OD600 values of the other samples (approximately 0.03 to 0.04).

4.1.6 Plasmid designing

***pac*-gene**

The *pac*-gene in the initial plasmid (*pac_old*), that should have been taken for the transformation via conjugation of the archaea, showed a different aminoacids (AA) sequence to the original sequence (*pac_org*) from *Streptomyces alboniger* (see (a) in fig. 4.10). They differ in five AA on the C-terminal end.
These mutation lid in to different 3-D structures. In (b) of fig. 4.10 the original *pac* has an

(a)

MTEYKPTVRLATRDDVPRAVRTLAAAFADYPATRHTVDPDRHIERVTELQ
ELFLTRVGLDIGKVWVADDGAAVAVWTTPESVEAGAVFAEIGPRMAELSG
SRLAAQQQMEGLLAPHRPKEPAWFLATVGVSPDHQGKGLGSAVVLPGVE
AAERAGVPAFLETSAPRNLPFYERLGFTVTADVEVPEGPRTWCMTRKPGA

pac_org

MTEYKPTVRLATRDDVPRAVRTLAAAFADYPATRHTVDPDRHIERVTELQ
ELFLTRVGLDIGKVWVADDGAAVAVWTTPESVEAGAVFAEIGPRMAELSG
SRLAAQQQMEGLLAPHRPKEPAWFLATVGVSPDHQGKGLGSAVVLPGVE
AAERAGVPAFLETSAPRNLPFYERLGFTVTADVECPKDRATWCMTRKPGA

pac_old

(b) **(c)**

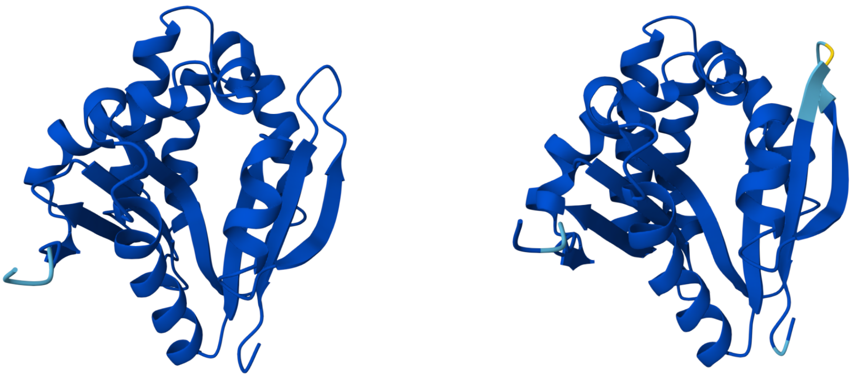

Figure 4.10: Aminoacids (AA) sequence of two *pac* genes and their 3-D structures. (a) shows the AA sequence of the original sequence from *Streptomyces alboniger* (*pac_org*) as well as from the *pac* gene that was located in the plasmid originally intended for transformation (*pac_old*). The differing section is highlighted in red. (b) The projected 3D structure for *pac_org* calculated by AlphaFold. (c) The projected 3D structure for *pac_old* calculated by AlphaFold.

antiparallel β-sheet with a big loop on the right side of its structure. The *pac_old* ((c)) shows also an antiparallel β-sheet at this position, but it is longer and has a shorter loop.

Plasmid

In SnapGene, four plasmid maps were created for *M. thaueri* and two for *M. boviskoreani*. The constructs followed the general scheme shown in fig. 1.1 and are listed in **??**. For the plasmids, the original *pac* sequence (*pac_opt*) was codon-optimized for the respective archaea.

Two plasmids were constructed, as shown in fig. 4.11: pMt_0231_hpt for *M. thaueri* and pMb_0231_hpt for *M. boviskoreani*. Compared to the plasmid originally used for the construction, a new restricitionsite (SpeI) was inserted between the *pac* and its promoter (see fig. 4.11). In addition, the plasmid *pac_old*, which contained the alternative AA sequence, was converted to the codon-optimized version with the original AA sequence (*pac_opt*), and

Table 4.3: Suicide plasmid constructs for transformation of *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani AbM4* (*M. boviskoreani*) via conjugation.The listed plasmids follow the general structure shown in fig. 1.1, with a cassette for *E. coli* containing the conjugation gene and homologous arms for recombination. The name of each plasmid is derived from the organism for which it possesses the appropriate homology arms (pMt for *M. thaueri* and pMb for *M. boviskoreani*) and the four named columns, where ORI refers to *M.* in this case and the promotor for the *pac*. *pac_org* is the *pac* with the original sequence derived from *Streptomyces alboniger*, *pac_opt* is the codon optimized version for the affected organism. The size of the constructed is given in bp as well.

name	ORI	promoter	pac	extra feature	size (bp)
pMt_0231_hpt	/ (0)	hmtB (2)	pac_opt (3)	hpt (1)	6145
pMt_0331_hpt	/ (0)	PglnA (3)	pac_opt (3)	hpt (1)	6254
pMt_0241_hpt	/ (0)	hmtB (2)	pac_org (4)	hpt (1)	6145
pMt_0341_hpt	/ (0)	PglnA (3)	pac_org (4)	hpt (1)	6254
pMb_0231_hpt	/ (0)	hmtB (2)	pac_opt (3)	hpt (1)	6214
pMb_0331_hpt	/ (0)	PglnA (3)	pac_opt (3)	hpt (1)	6323

the promoter was changed to hmtB.

Both plasmids were verified by sequencing.

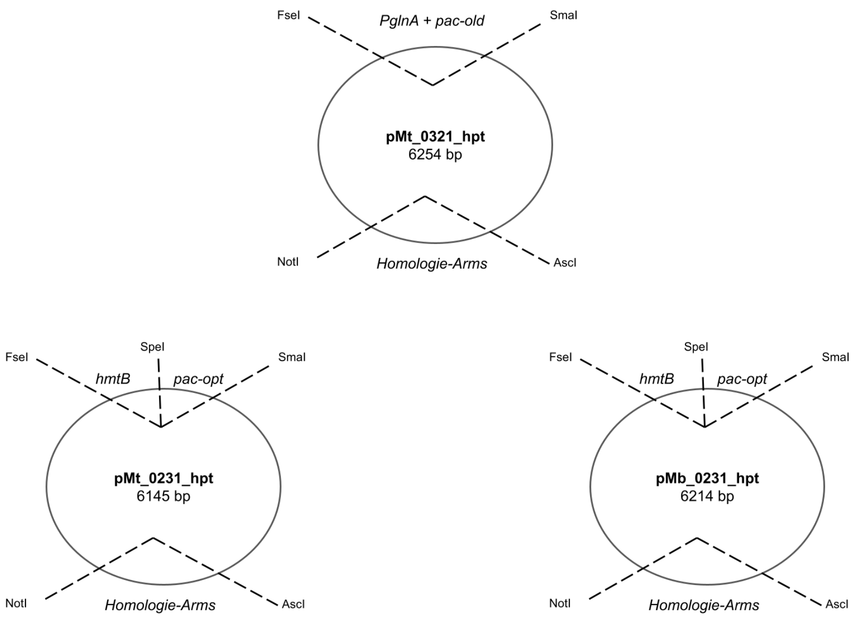

Figure 4.11: Suicide plasmid maps for transformation by conjugation for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani AbM4* (*M. boviskoreani*). pMt_0321_hpt is the parent plasmid used to construct the other two. The maps show only the gene regions that have been modified, as well as the restricitionsite sites used or the newly created restricitionsite sites.

4.1.7 Transformation via conjugation

***E. coli* cultivation test in DSM 1523**

Cultivation of *E. coli* $DH5\alpha$ was tested under similar settings in DSM 1523 as for *Methanobre-*

vibacter. Three drops for inoculation was enough to get a dense culture the next day.

Preparation for transformation

Hier erst in Leipzig text möglich, weil da das laborbuch liegt -> erstelle Tabelle mit Ansätzen und dann welche erfolgreich waren

+ Sequenzergebnisse -> bestätigung von bovis ja vorhanden -> komplettes genom is still pending

4.2 Geography

All methane emission analyses focused on three countries: Germany (DEU), New Zealand (NZL) and United States of America (USA). In addition, only the anthropogenic emission will be investigated.

4.2.1 Total methane emissions of a country

Figure 4.12 shows the development of the total methane emission of a year as well as of the specific sectors from 1990 to 2020s for each country.

In general, the first thing that stands out is that the time series for the U.S. ends as early as 2022, while those for the other two countries do not end until 2024. Furthermore, the scale of methane emissions shows that the U.S. generally produces significantly more CH_4; the maximum total emissions are 35,000 kt CH_4, while for DEU the maximum is 5 Tg and for NZL it is 1.5 Tg.

For Germany (DEU), a clear downward trend is evident in total CH_4 emissions (see fig. 4.12). Emissions fell from 3 Tg CH_4 in 1990 (5 Tg) to 2 Tg in 2024, representing a decrease of approximately 60%. Two distinct phases of this trend can be identified: first, from 1990 to 2006, with a sharp decline of about 2.5 Tg CH_4 (approximately 2.6 Tg CH_4 in 2006), followed by a phase with a more moderate decline to 2 Tg of anthropogenically produced methane in 2024.

Total emissions excluding those from enteric fermentation (EnterFe) in cattle (total excluding cattle EnterFe) follow a similar trend, with the curve's value being only 1 Tg CH_4 lower.

Looking at the sectoral emissions of DEU, it becomes apparent that in 1990 there were three equally large CH_4-producing sectors: "Energy," "Waste," and "Agriculture," each accounting for approximately 1.5 Tg CH_4 (see fig. 4.12). However, these sectors have developed differently.

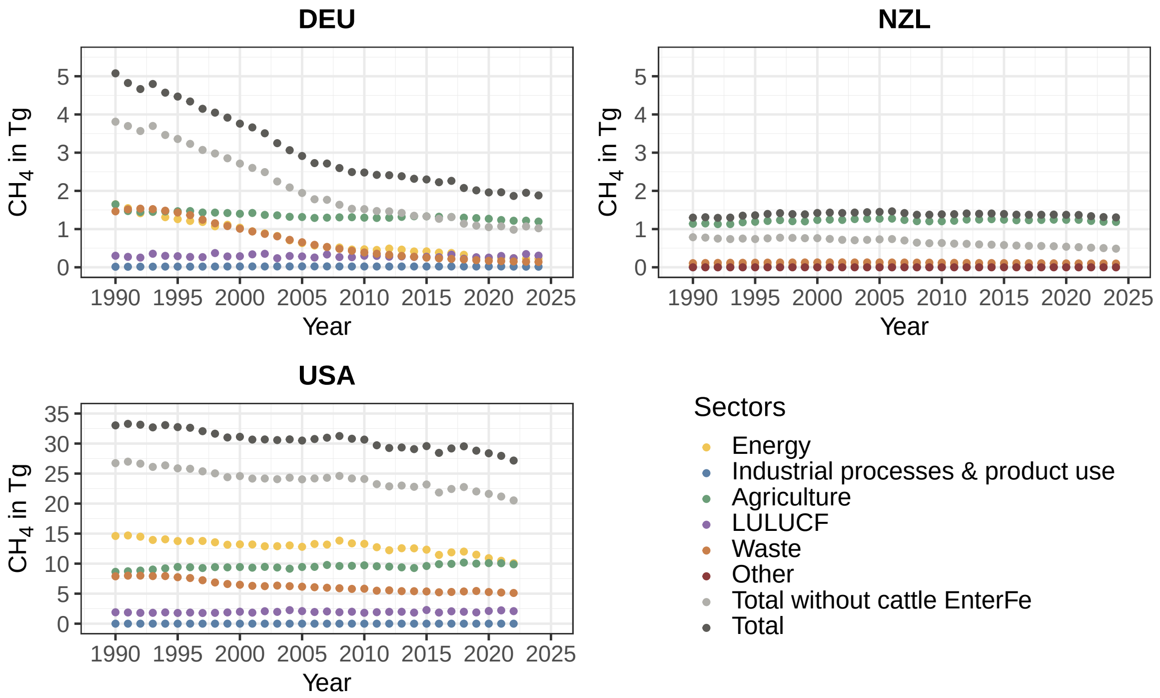

Figure 4.12: Time series of the anthropogenic methane emission of Germany (DEU), New Zealand (NZL) and United States of America (USA). The different relevant emission sectors are shown with their total sum (Total). Furthermore, the total emission minus the CH_4 produced in enteric fermentation (EnterFe) processes of cattle is plotted as well. Methane is plotted as Tg CH_4 of a year against the time.

The energy and waste sectors have followed quite similar trends. Following a sharp decline between 1990 and 2006—with the waste sector not beginning to change until 1994—from 1.5 to 0.5 Tg, a moderate decline then sets in, until approximately 0.025 Tg of CH_4 is emitted in 2024.

Agriculture is also changing, though only very slightly compared to the other two sectors: from 1.6 Tg of methane in 1990 to approximately 1.25 Tg CH_4 in 2024, with the largest single change occurring between 1990 and 1991, amounting to a difference of about 0.1 Tg. In 2015, total emissions excluding cattle EnterFe are exactly as high as those of the agricultural sector and fall below that level thereafter.

In addition to these three sectors, two others are shown in fig. 4.12.

The “Industrial Processes & Product Use” sector is a negligibly small source of methane emissions. Its CH_4 emissions have remained consistently well below 0.1 Tg over the 35-year period.

With CH_4 emissions consistently ranging between 0.3 and 0.4 Tg, the land use and land use change and forestry (LULUCF) sector was the fourth-largest single methane emission sector in Germany during the 1990s, but it surpassed the waste sector in 2016 and has been the second-largest CH_4-emitting sector in DEU after agriculture since 2019.

For the USA the difference between total methane emissions and total emissions excluding cattle EnterFe remains fairly constant at around 10 to 15 Tg CH_4 (see fig. 4.12). In addition, a clear, steady decline can be observed for the total methane emission from 1990 (approx. 33 Tg CH_4) to 2022 (approx. 27 Tg CH_4), although the curve does not follow the trend linearly but rather fluctuates slightly. Over this 32-year period, the U.S. has reduced its methane emissions across all sectors by about 6 Tg CH_4, which corresponds to a reduction of 18%.

As in Germany, the three most significant methane-producing sectors in the U.S. are "Energy," "Agriculture," and "Waste" in 1990; however, here the energy sector is the largest contributor, accounting for nearly half of the emissions in 1990 with 15 Tg CH_4, followed by the waste and agriculture sectors with approximately 8 Tg CH_4 each.
The energy and waste sectors show a downward trend over the time series (energy: from 15 to 10 Tg CH_4 and waste: from 8 to 5 Tg CH_4), while emissions from agriculture rise slightly over time, from 8.5 Tg to 10 Tg CH_4. Thus, in 2022, agriculture and energy—each at 10 Tg CH_4—are the largest methane emission sectors and together account for more than two-thirds of total methane emissions in the U.S. in 2022.
Two other sectors contribute to U.S. methane emissions: the "Industrial Processes & Product Use" sector, which is relatively small and therefore negligible, and the "LULUCF" sector, with a constant range of 0.2 to 0.25 Tg CH_4.

Compared to the other two countries, New Zealand presents a different picture (see fig. 4.12). First, total CH_4emissions in 2024 are nearly the same as those in 1990, namely about 1.4 Tg CH_4; furthermore, a slight increase in emissions to 1.5 Tg CH_4 can be observed through 2006, followed by a slight decline.
Second, "agriculture" is the dominant sector in terms of methane emissions and accounts for nearly 90% of CH_4 emissions from NZL (see fig. 4.12). All other sectors account for significantly less than 0.1 Tg CH_4 per year, with the exception of the waste sector, whose methane emissions are slightly above 0.1 Tg CH_4.
Third, the difference between total methane emissions and total emissions excluding cattle EnterFe increases by about 60% (0.5 Tg CH_4 in 1990 and 0.8 Tg in 2024).

4.2.2 *Enteric fermentation processes of cattle*

Methane emissions

Figure 4.13 shows the trend in methane emissions from enteric fermentation in cattle, as well as their subcategories “dairy cattle” and “non-dairy cattle” for each country. The values for the subcategories are calculated according to eq. (1) and based on the required data from the reported CRT (visualized in fig. A2 through A4). Total emissions from cattle are therefore simply the sum of these values.

Cattle emissions for Germany (DEU) in fig. 4.13 decline from 1990 to 2024 (from 1.25 Tg CH_4 to approximately 0.8 Tg), with a steeper decline in the first year of the time series, followed by a more gradual decline.
For dairy and non-dairy cattle, this initial decline is also evident in fig. 4.13, after which a slight decline can also be observed (dairy cattle: 0.75 Tg CH_4 in 1990 to 0.5 Tg in 2024; non-dairy cattle: 0.55 in 1990 to 0.35 in 2024). Dairy cattle are an exception: their emissions drop to 0.5 Tg CH_4 by 2006, then rise slightly again through 2015, and subsequently decline once more, returning to 0.5 Tg by 2014.

Dairy cattle consistently produce slightly more CH_4 per year than non-dairy cattle (0.1–0.2 Tg CH_4, see fig. 4.13. This contrasts with the herd ratio shown in fig. A2, where the dairy cattle herd generally accounts for about half of the non-dairy cattle herd over the years.
Further differences in the indicators are evident for Ym, where dairy cattle had a significantly higher value of 7% in 1990 compared to 6.25, which, however, declines sharply to 6.1% by 2024, while the value for non-dairy cattle rises to 6.5%, as well as for GE, where the value for the non-dairy cattle herd remains constant at 100 MJ $head^{-1}$ day^{-1}, while the value for the dairy cattle herd rises from 250 to 380 MJ $head^{-1}$ day^{-1}.

Methane emissions from cattle in EnterFe for New Zealand rise from 0.5 to 0.75 Tg CH_4, while the values have remained relatively constant since 2014.
An increase is also evident for dairy cattle, which has plateaued since 2014 (0.25 Tg CH_4 in 1990 to 0.55 Tg in 2024). Non-dairy cattle, on the other hand, show similar values of around 0.5 Tg CH_4 in both 1990 and 2024.
Thus, in 1990, dairy and non-dairy cattle contributed equally to methane emissions from the cattle population; however, this share has changed over the years for dairy cows, so that by 2024 they will account for about two-thirds of the emissions. This shift in the ratio is quite

similar across the subcategory populations (see fig. A3). Furthermore, the energy efficiency (GE) values for non-dairy cattle remain constant at 150 MJ head^{-1} day^{-1}, while the value for dairy cattle rises from 160 to 225 MJ head^{-1} day^{-1}.

In the USA, methane emissions from dairy cattle rise from just under 2 Tg CH_4 to 2 Tg, with the slight increase not beginning until around 2005. The other two livestock categories, on the other hand, fluctuate overall—for both non-dairy and dairy herds—between 4.9 and 5.5 Tg for non-dairy livestock and between 6.5 Tg CH_4 and 7.1 Tg for dairy livestock (see fig. 4.13). Unlike in the other two countries, non-dairy cattle here emit significantly more than dairy cattle (2024: 5 Tg CH_4 versus 2 Tg).

As with the other countries, the emission curves correspond fairly well with the livestock numbers from fig. A4, where non-dairy cattle—at 80 million head—are kept in significantly greater numbers than dairy cattle (20 million head). The curve for dairy cattle does not match the livestock numbers exactly, as these remain constant. However, the GE for dairy cattle rises steadily from 200 to 260 MJ/animal·day between 1990 and 2022, while the Ym value decreases from 6.5% to 6.1% over the same period.

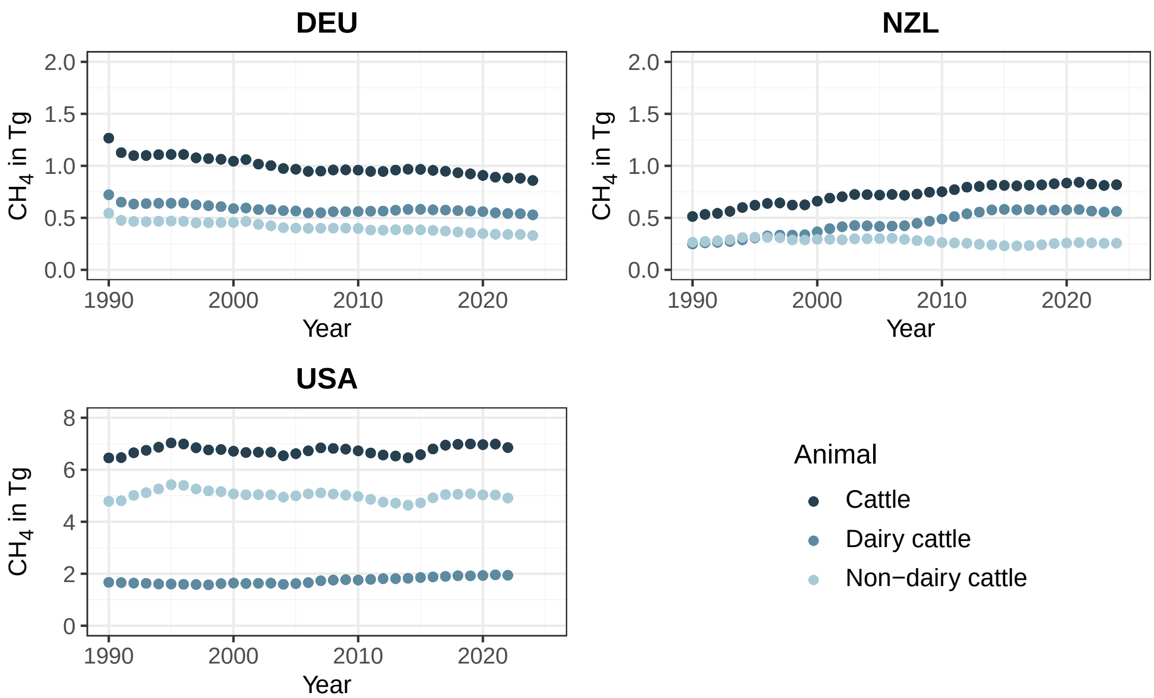

Figure 4.13: Time series of methane emission derived from enteric fermentation processes in cattle and its subcategories dairy and non-dairy for the countries: Germany (DEU), New Zealand (NZL) and United States of America (USA). Methane emission are measured in emitted Tg CH_4. Emissions are alculated for dairy and non-dairy cattle with the indicators in fig. A2:A4 and the eq. (1). The cattle emission is than the sum of its subcategories.

When comparing cattle emissions across countries in fig. 4.13, it is striking that the U.S. produced more than three times as much in 2022 as the other two countries combined (6.8 Tg CH_4 compared to less than 2 Tg combined). This is also reflected in the number of animals: in 2022, the US had approximately 100 million head of cattle, while DEU and NZL each had around 10 million (see fig. A2:A4).
Germany and New Zealand produce similar amounts of methane from cattle farming in 2024 (approx. 0.8 Tg CH_4), in 1990, DEU had produced twice as much, but also kept twice as many cattle (approx. 20 million animals in 1990 compared to 10 million in 2024).

Share of CH_4 emissions from EnterFe of cattle

The share of CH_4 produced through EnterFe processes in cattle on entire amount of emitted methane for the agriculture sector of a country declines slightly for Germany (approx. from 78% 1990 to 76% 2024, see fig. 4.14) and USA (from 76% 1990 to 74% 2022, see ibid.). For the share on total methane emissions, both curves increase over time quite linear, but is the increase fur the US mild from 20% (1990) to 25% (2024), is it for DEU quit intense from 1990 also a value of around 20-25% to 2024 with 45%.

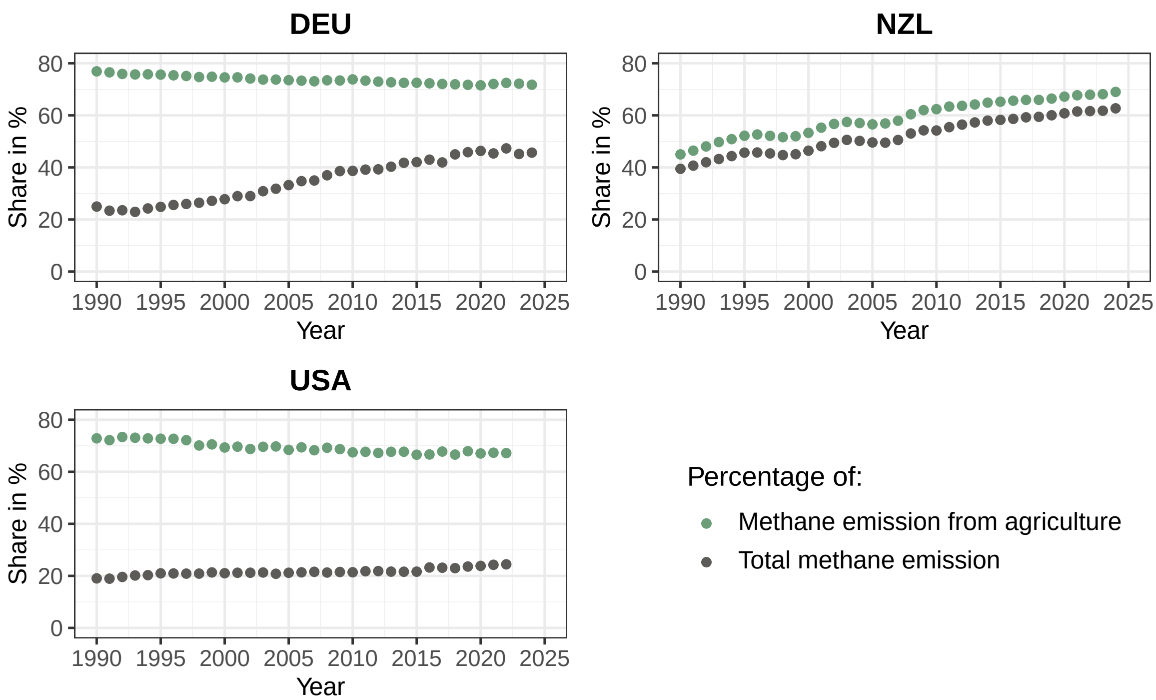

Figure 4.14: Time series of the share of methane emissions resulting from enteric fermentation of cattle on methane produced in the entire agriculture sector and the total amount of CH_4 per country and year. The calculation of the share are based on the information of the reported CRT.

New Zealand time series for the development of the share of enteric fermentation cattle CH_4

emission looks a bit different to those described (see fig. 4.14). Both curves for the share on total and agriculture methane emission show a similar form, they just have slightly different values. The share on agriculture on the one hand grew from 1990 to 2024 from 45% to 70%, on the other hand the share for total emission from 40% to a bit less than 65%.

4.2.3 Reductionpotential in enteric fermentation processes of cattle

Hier muss definitiv in der Disskusion ein Vergleich mit der atmospheric growth rate von 20.1-21.7 Tg CH_4 / year dür diese dekade kommen -> in sangios et al/ siehe präsi + kann hier auch darüber sprechen, dass linearität von Ym eindeutig den von DE überwiegt

Figure 4.15 shows the average reduction in CH_4 emissions during the decade from 2010 to 2019 in the countries NZL, DEU, and USA for various possible scenarios in which the shift in the rumen microbiome was successful to varying degrees. The scenarios are 10, 20, 30, 40, and 50, where the numbers refer to the percentage by which the microbiome is shifted toward non-methane-producing microbes. Table table B8 in the appendix lists the precisely calculated amounts of saved CO_2 eq. and CH_4.

For the various scenarios, the CH_4 diagram shows that, under identical scenarios, by far the largest amount of CH_4 can be saved in the United States, and that Germany and New Zealand make roughly equal contributions over the ten-year period examined, with Germany contributing slightly more. Overall, a 10% shift would result in annual emissions of approximately 0.9 to 1 Tg CH_4 less over the decade under consideration, with about 0.75 Tg attributable to the United States alone.
At 30%, the total reduction in methane emissions would be approximately 2.75, and at 50%, it would be almost exactly 4.5 Tg. Table B8 However, the appendix shows that the reductions are not entirely linear but follow a slightly nonlinear trend, even though this is depicted differently in fig. 4.15 (for example, DEU reaches 104.40 Tg with a 10% shift and 206.52 Tg CH_4 with a 20% shift).

For global warming potential (GWP), significant quantitative differences can be observed depending on the amount of gas emitted. For example, in a “scenario with a 30% emission reduction” over a 20-year time horizon, approximately 220 Tg less eq. CO_2 would be emitted across all countries combined; over a 100-year time horizon, this figure would be only 75 Tg. The relative dominance of the United States remains evident in these figures.

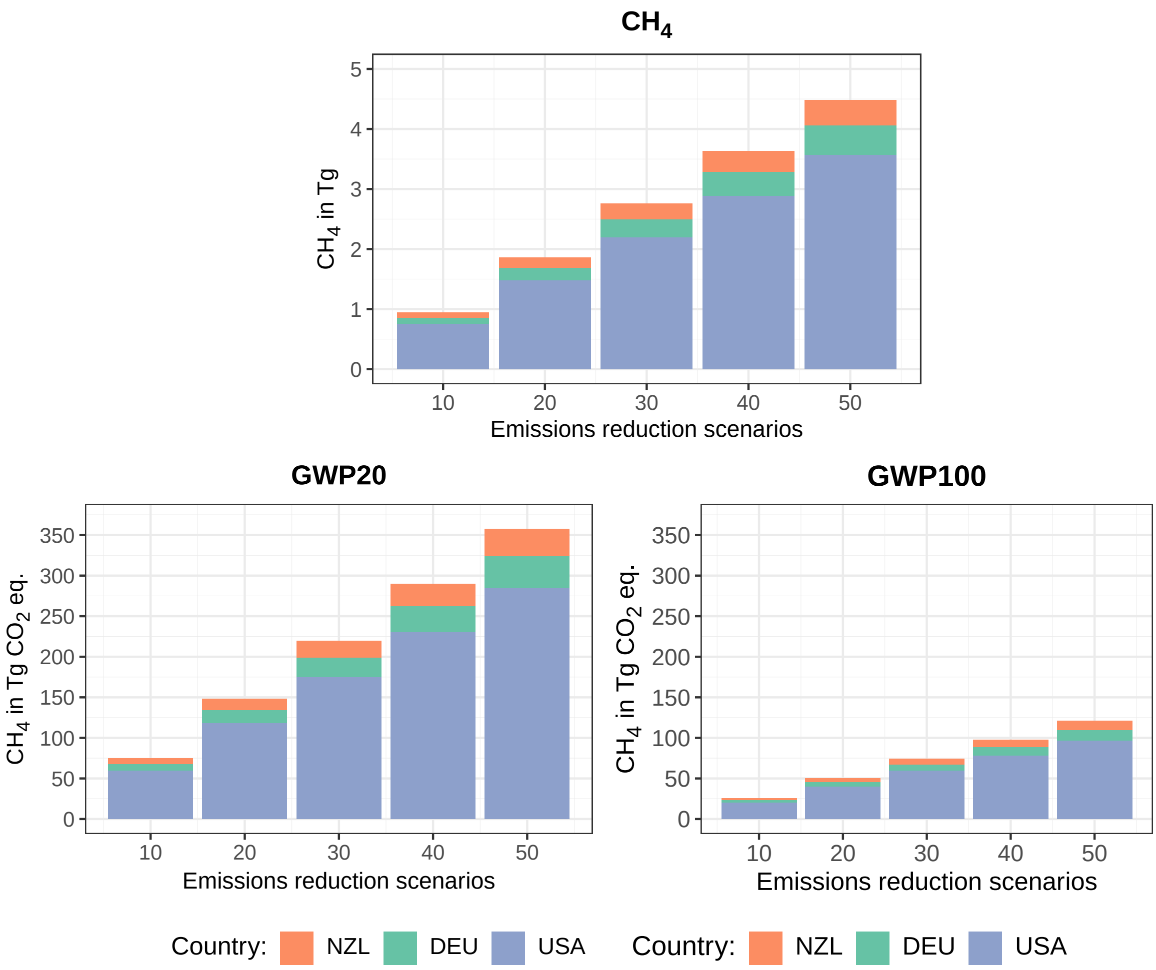

Figure 4.15: Saved CH_4 through different scenarios (10, 20, 30, 40 and 50%) of a successful shift of rumen microbiome towards less methanogenic microbes. The saved methane is plotted as Tg CH_4 against the these five "Emission reduction scenarios" (CH_4). The values for the global warming potential (GWP) for 20 and 100 years are plotted as well. The saved CH_4 is calculated as an average for the decade 2010-2019 for each country (Germany (DEU), New Zealand (NZL) and United States of America (USA))

4.2.4 Prediction for 2030

Figure 4.16 (DEU), 4.17 (NZL) and 4.18 (USA) show the total methane emission for each country as a time series from 1990 to the last published year in the CRT and beyond until 2030 as a forecast. The forecast is shown for three different linear models (using 5, 10, or 15 years as the reference period for the fit) and for a scenario in which no further intervention was made, as well as for three scenarios with varying degrees of success in cattle immunization (10, 30, and 50% shifts in the rumen microbiome). The equations as well as the R^2 and confidence intervall (CI) values are given in the annex in

A similar trend is in all three figures visible for the three linear fits used. All models show a negative developement, although the max. confidence intervall (CI) of the 5 year model has a positive trend.

Besides, the forecast for the fit with 10 or 15 years as time reference show very similar prediction while the 5-year linear model differs from them. But in DEU the 5-year fit predicts higher values for CH_4 production for 2030 than the other two models, while for NZL and USA the fits with the longer time sample as reference higher values predicts.

Two additional points stand out when comparing Germany with the other two countries (fig. 4.16 to fig. 4.17 and 4.18).

First, in Germany, the predicted range of emission values at least partially meets the two emission targets set by the global methane pledge (GMP), whereas in the other two countries, the emission values exceed the targets.

Second, the value for the 60% of 2010 emissions target is higher than the 70% of 2020 emissions target for DEU. For the other two countries, it is the other way around. Even so, the two values are in a similar range for all of them.

A closer look at the values in fig. 4.16 for Germany shows that the five-year model for predicting the emissions trend, assuming unchanged mitigation efforts, forecasts a nearly constant trend with a slight decline (from 1.9 Tg CH_4 to 1.8 Tg), although the confidence intervals here are quite wide (see **??** from X to Y), and an increase in the values is even possible here.

The 10- and 15-year models, on the other hand, show a clear downward trend with a value of approximately 1.5 Tg CH_4 in 2030, which corresponds almost exactly to the 60% target of the GMP. The range defined by the CI is also significantly narrower here, ranging from about 1.4 to 1.75 Tg, with the 70%-target remaining just above the lower CI range.

The cumulative range of all three confidence intervall values points to a range of approximately 1.4 Tg to 2.25 Tg CH_4.

The projected values for the reduction scenario 10 hardly differs from the scenario without mitigation measures and deviates by only about 0.1 Tg (see fig. 4.16).

A greater difference can be observed for Scenario 30. Here, the projected emissions for the 5-year adjustment would be roughly at the level of the GMP target of 60% of 2010 emissions and thus at values comparable to those in the 10- and 15-year models under “Projection without vaccination.” For this scenario, the values for these two models under “with immunization” are just below the target of 70% of 2020 emissions, namely at approximately 1.3 Tg CH_4. The range of cumulative CI for the year 2030 extends from 1.1 Tg to just under 2 Tg of emitted CH_4. This means, for these models assuming a 30% change in the rumen microbiome, no

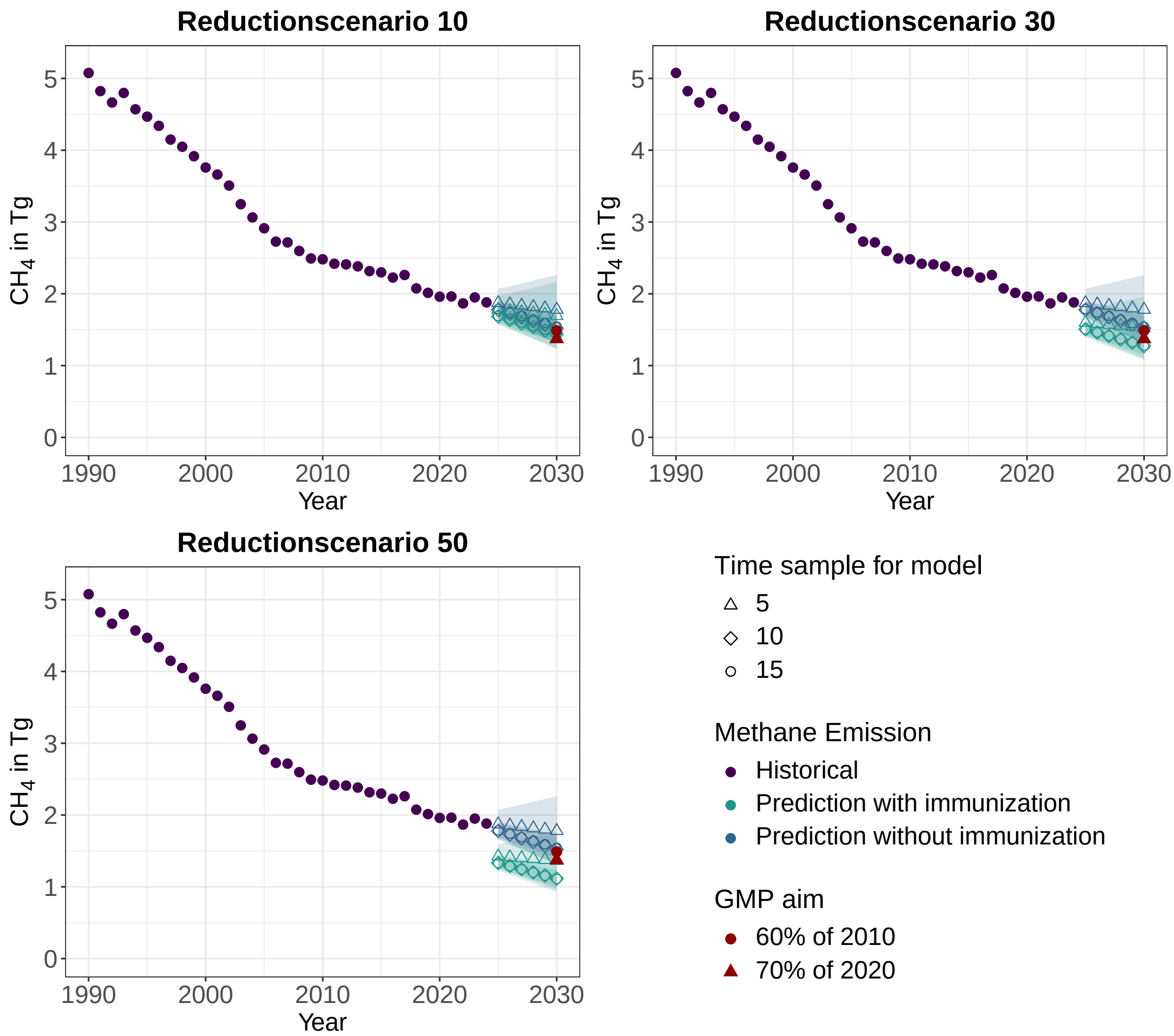

Figure 4.16: Historical values for total methane emissions in Tg for **Germany** from 1990 to 2024 and the forecast of methane emissions through 2030. The projection was generated using three different linear models with varying time reference periods (the last 5, 10, or 15 years), which are listed in **??** along with the corresponding equation and R^2. Total emissions were not extrapolated directly. Instead, they were determined using a prediction of total emissions excluding enteric fermentation in cattle, as well as the four indicators (DE, GE, NE_x, and herd size), to calculate the missing emissions from cattle production. In addition to predicting total CH_4 emissions without additional mitigation measures (blue), the calculation was also performed for three reduction scenarios (10, 30, and 50% shift in the rumen microbiome) (green). For each linear extrapolation, the confidence intervall are displayed as a colored area in the corresponding color (confidence level of 95% was used). The two global methane pledge (GMP) methane emission targets are also shown in red: 60% of 2010 methane emissions as a circle and 70% of 2020 methane emissions as a triangle.

further increase in methane levels would be expected until 2030.

The scenario 50 with an immunisation of cattle that leads to a 50% shift of microbiome towards less methan production, had an combined CI range from slightly below 1 Tg to 1.75 Tg CH_4 emitted in 2030. Where the models and their CIs would be completly below both GMP emission targets for 2030 (see fig. 4.16, approx. slightly below 1 Tg to 1.3 Tg CH_4).

In addition, the prediction for the 5-year linear fit would be 2030 at nearly exact the 70% emissions of 2020 (around 1.4 Tg CH_4), although the max. CI value is with 1.75 Tg clearly above both emission aims.

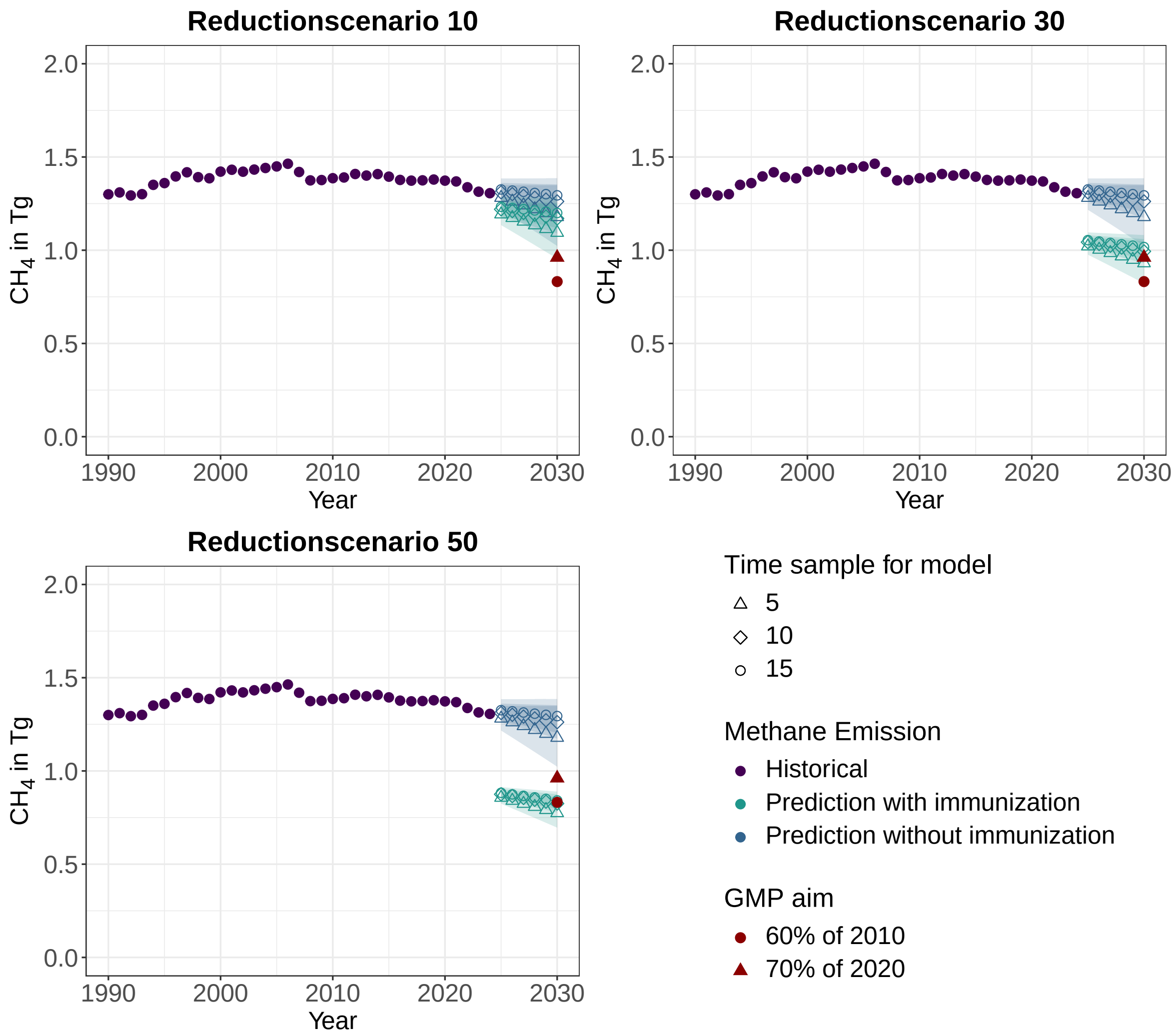

Figure 4.17: Historical values for total methane emissions in Tg for **New Zealand** from 1990 to 2024 and the forecast of methane emissions through 2030. The projection was generated using three different linear models with varying time reference periods (the last 5, 10, or 15 years), which are listed in **??** along with the corresponding equation and R^2. Total emissions were not extrapolated directly. Instead, they were determined using a prediction of total emissions excluding enteric fermentation in cattle, as well as the four indicators (DE, GE, NE_x, and herd size), to calculate the missing emissions from cattle production. In addition to predicting total CH_4 emissions without additional mitigation measures (blue), the calculation was also performed for three reduction scenarios (10, 30, and 50% shift in the rumen microbiome) (green). For each linear extrapolation, the confidence intervall are displayed as a colored area in the corresponding color (confidence level of 95% was used). The two global methane pledge (GMP) methane emission targets are also shown in red: 60% of 2010 methane emissions as a circle and 70% of 2020 methane emissions as a triangle.

Figure 4.17 shows the prediction for NZL. The prediction without immunization has a combined CI range from slightly above 1 Tg to 1.35 Tg emitted methane for 2030. Here, the CI is as before for DEU notable wider for the 5-year model as the other two. Neither the values directly predicted through the models (1.2 Tg CH_4 for the 5 year fit and 1.25 to 1.3 Tg for the 10- and 15-year model) nor the CI range achieves one of the two GMP emission targets. The 60% of 2010 emissions is at 0.8 Tg and 70% of 2020 at slightly low 1 Tg CH_4.

The difference between no immunization and immunization for the reduction scenario 10 is like for Germany in fig. 4.16 round about 0.1 Tg, but this is enough that the minimum of the CI to be low enough to achieve just the 70% target (see fig. 4.17).
At the reduction scenario 30, this aim lays between the prediction of th 5-year (approx. 0.9 Tg CH_4) and the etrapolation with the 10- and 15-year model (1 Tg). Besides, the lower limit for the confidence intervall reaches the other GMP aim.
For the last assumed scenario (50% shift of microbiome) the 10- and 15-year linear model predict methane emission values that are nearly exact the 60% target. So, most of the predicted reange for the methane emission achieves this GMP aim with an range from 0.7 Tg to 0.85 Tg emitted methane for the year 2030.

For the U.S., the 10- and 15-year models for "prediction without immunization" in fig. 4.18 predict an emissions value for 2030 that corresponds almost exactly to the 2022 value (approximately 27 Tg CH_4). The 5-year fit, on the other hand, shows a significant decrease by 2030—a reduction of approximately 3 Tg to 24 Tg of methane emitted in 2030. Even with the combined CI range (22 to 28 Tg CH_4), the lowest value is still significantly higher than the nearest GMP emissions target (20 Tg CH_4 for the 70% target).

Even the confidence intervall range of reduction scenario 10, which differs by approximately 1 Tg from the scenario with no additional mitigation efforts, does not meet this target with its lowest value of approximately 21 Tg.

With reduction scenario 30, the prediction for the 5-year model is still slightly above the target at 21 Tg CH_4, but the CI range falls just below 20 Tg and thus below the 70% methane emissions level of 2020. The maximum value for the CI range is approximately 26 Tg, meaning that under this scenario, no values predicted for the year 2030 are above the 2022 value (see fig. 4.18).

In the 50% reduction scenario, the 5-year model prediction, with a value of approximately

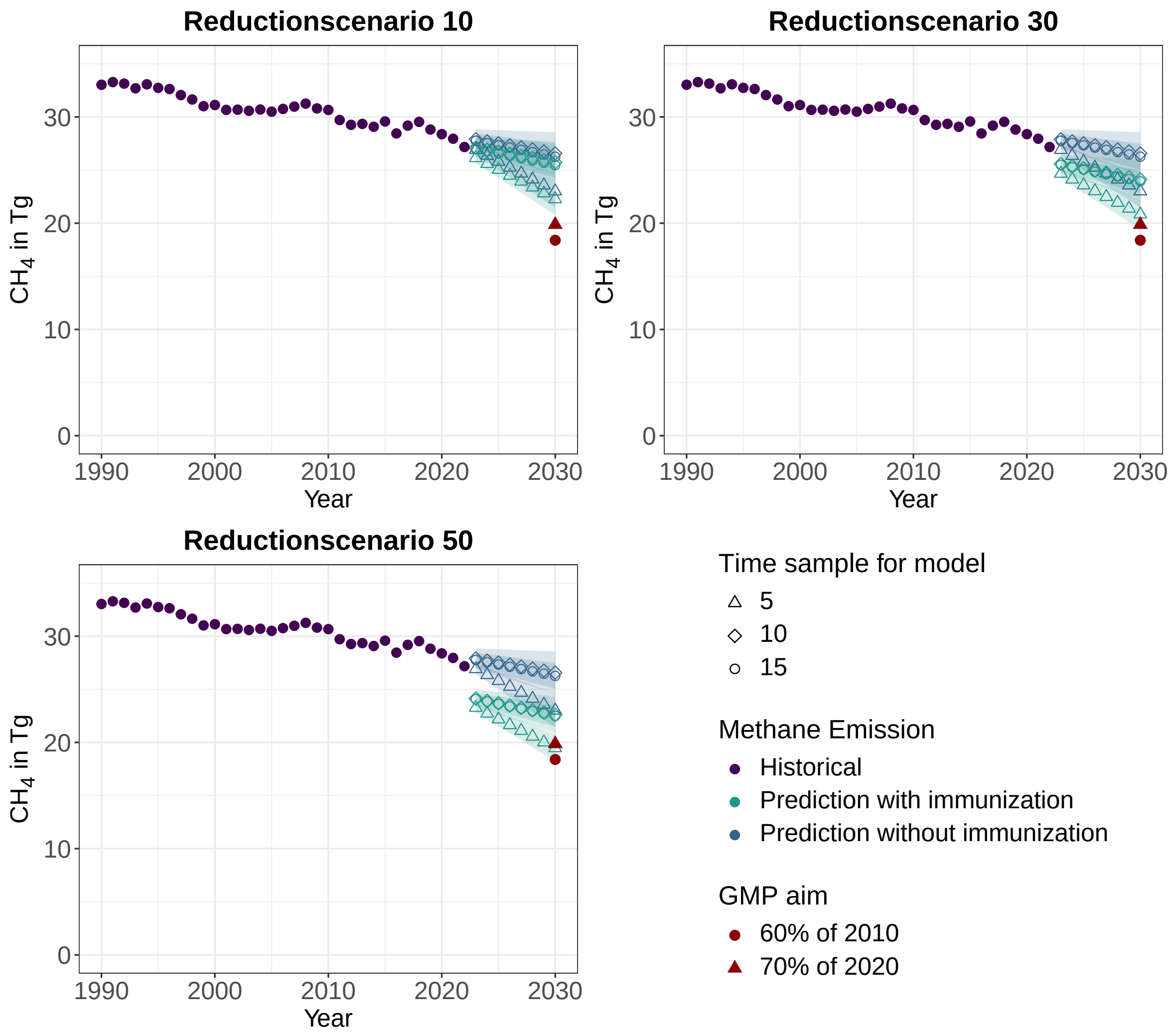

Figure 4.18: Historical values for total methane emissions in Tg for the **USA** from 1990 to 2022 and the forecast of methane emissions through 2030. The projection was generated using three different linear models with varying time reference periods (the last 5, 10, or 15 years), which are listed in **??** along with the corresponding equation and R^2. Total emissions were not extrapolated directly. Instead, they were determined using a prediction of total emissions excluding enteric fermentation in cattle, as well as the four indicators (DE, GE, NE_x, and herd size), to calculate the missing emissions from cattle production. In addition to predicting total CH_4 emissions without additional mitigation measures (blue), the calculation was also performed for three reduction scenarios (10, 30, and 50% shift in the rumen microbiome) (green). For each linear extrapolation, the confidence intervall are displayed as a colored area in the corresponding color (confidence level of 95% was used). The two global methane pledge (GMP) methane emission targets are also shown in red: 60% of 2010 methane emissions as a circle and 70% of 2020 methane emissions as a triangle.

19.5 $TgCH_4$, is slightly below the 70% target but still above the GMP's 60% target relative to 2010 emissions (approximately 18 Tg CH_4). However, the CI range of the 5-year fit (18 to just under 21 Tg CH_4) meets this target.

The other fits, as well as their CI range (21 to 24 Tg CH_4), do not meet either of the two GMP targets even in reduction scenario 50 (see fig. 4.18).

5 DISCUSSION

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6 CONCLUSION

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BIOCHEMICAL TERMS

β-sheet	A common secondary structure of proteins. It consists of β-strands arranged in parallel or antiparallel orientations. β-strands are illustrated as arrows. Another typical secondary structure is the α-helix.
16S-Sequencing	16S sequencing targets the 16s genes, which code for the 16S ribosomal RNA (rRNA). The 16S rRNA is a part of the procaryotic ribosome. These 16-s base sequenz is specific for the organism.
AlphaFold	Is a KI based software for prediction of 3-D structure of proteins. For the prediction it needs an aminoacid sequence.
archaea	Archaea constitute one of the three major domains of life (bacteria, archaea, eukaryotes), and, like bacteria, they belong to the prokaryotes (Fritsche, 2024, p. 3).
C-terminal end	The end of the amino acid sequence of a protein, where the carboxy group of the terminal amino acid has no covalent connection to another amino acid.
colony	In a microbiological context, a colony is defined as a clearly distinct cluster of cells that can be traced back to a single parent cell through cell division.
complex media	Complex media contain natrual, organic moleculas, that are not precisely defined chemically(e.g. yeast extract or pepton) So a lot of diffrent microbes can grow on them (Fritsche, 2024, p. 359).

glovebox	A container that is gas-tight, so that a different atmosphere can be created.
ORI	Origin of replication, is the region of the DNA, where the machinery responsible for replicating the entire genomic DNA first binds and begins replication.
phase-contrast microscope	Phase-contrast microscopy is based on the fact, that objects or medium not only changes the amplitude of light but also the phase of it. The phase-contrast microscope makes this phenomena visual for the human eye (Pawlizak, 2009).
plasmid	A small ring of double-stranded DNA. Occurs naturally in archaea and bacteria, but is also used for scientific purposes. Here often as a vector for a transformation.
Plating	Cultivate microorganism on a solid medium (contains a gelling agent like agar). THerby is the plating the process how the organism get distributed on the plate / solid media. Different technics are possible like spreading or pouring (Fritsche, 2024, p. 364).
pour plating	A plating technique in which the solution containing microorganisms is mixed with the agar-containing culture medium while it is still liquid (at about 45 °C). This mixture is then poured into a Petri dish, where it solidifies(Fritsche, 2024, p. 364).

promotor	Region on DNA, which allows binding of relevant enzyms for the transcription of genes. So it is a essential region for gene expression.
restricitionsite	Restrictionsites are a specific sequence of approx. 4 to 10 bp that are recognized by a specific restriction-enzyme. This cuts the double-stranded DNA at this site in a specific behavior.
spread plating	A plating technique in which the solution containing microorganisms is put on the solid agar-containing culture medium. The solution is then spread across the agar plate using a Dragalski spatula (Fritsche, 2024, p. 364).
transformation	The process of getting a extern DNA into the target organism and its expression. If the target organism is a animal eucaryotic cell it is called transfection.

GEOGRAPHICAL TERMS

CRT Common reporting table is a template for countries to report its emission of GHG fitting IPCC standards. The UNCFF has oblige the ANNEX-I countries to publish this every year since 1990.

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APPENDIX

A Figures

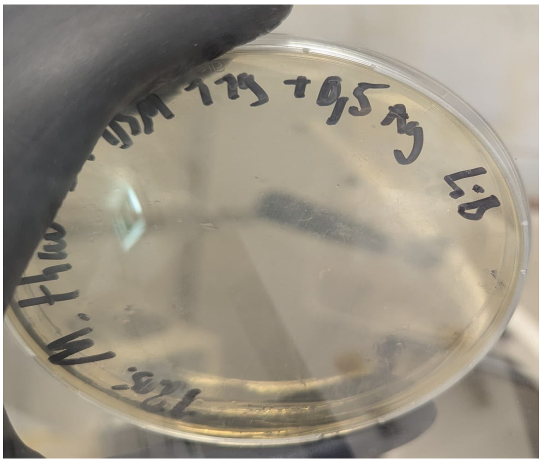

Figure A1: Pour plating result of *Methanobrevibacter thaueri CW*^{T} (*M. thaueri*) DSM 119 (1% agar). 100 µl of a full-grown culture have been plated. Cultivated in a jar with H_2/CO_2 gas phase. No growth detected.

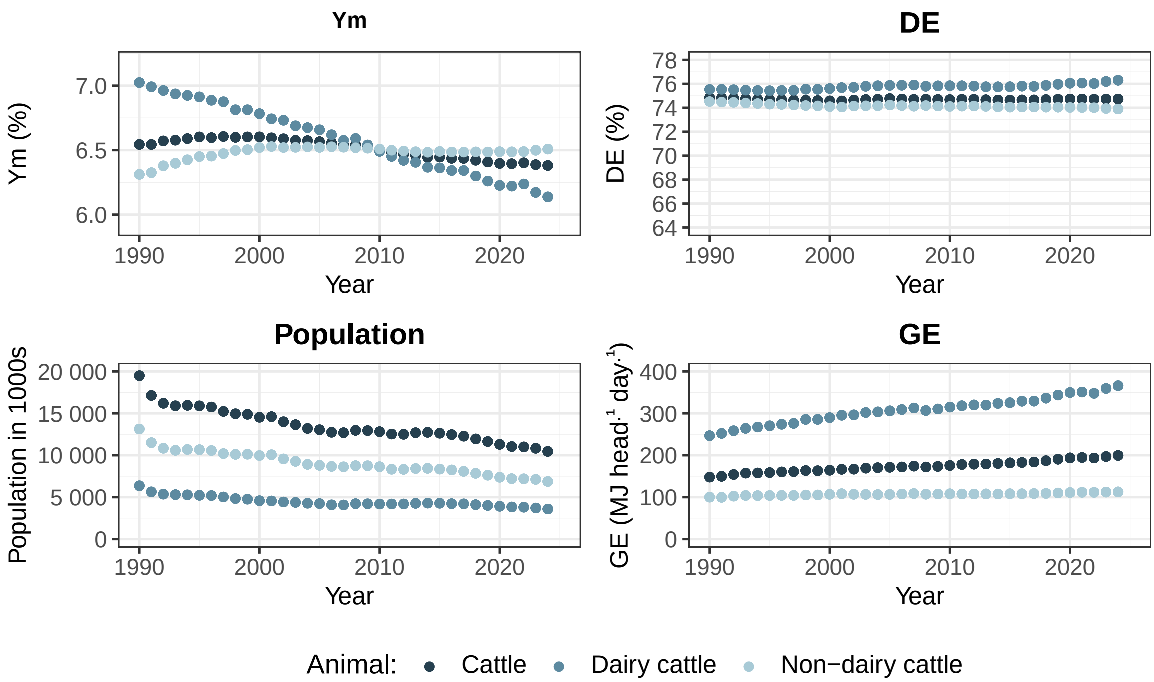

Figure A2: Time series of methane conversionfactor (Ym), digestibility of food (DE), population-size and gross energy intake (GE) of cattle and its subcategories dairy and non-dairy for Germany.The data for dairy and non-dairy cattle is from the published CRT. The values for cattle is calculated as weighted mean of its subcategories, where the population-size is used as weight.

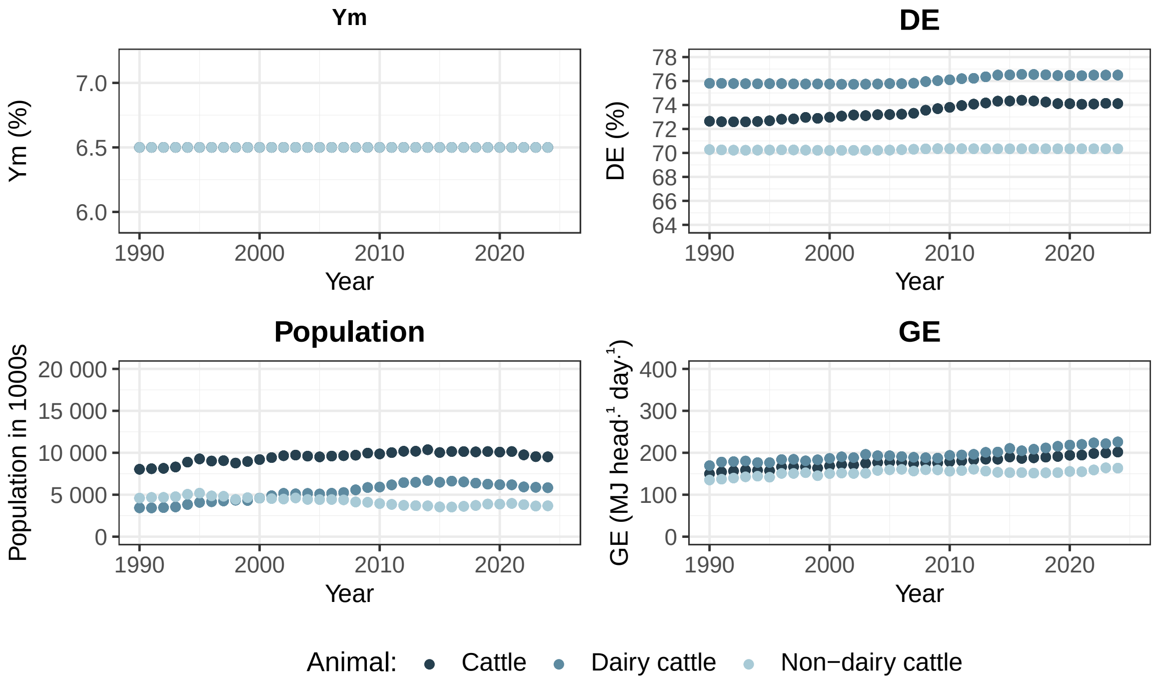

Figure A3: Time series of methane conversionfactor (Ym), digestibility of food (DE), population-size and gross energy intake (GE) of cattle and its subcategories dairy and non-dairy for New Zealand.The data for dairy and non-dairy cattle is from the published CRT. The values for cattle is calculated as weighted mean of its subcategories, where the population-size is used as weight.

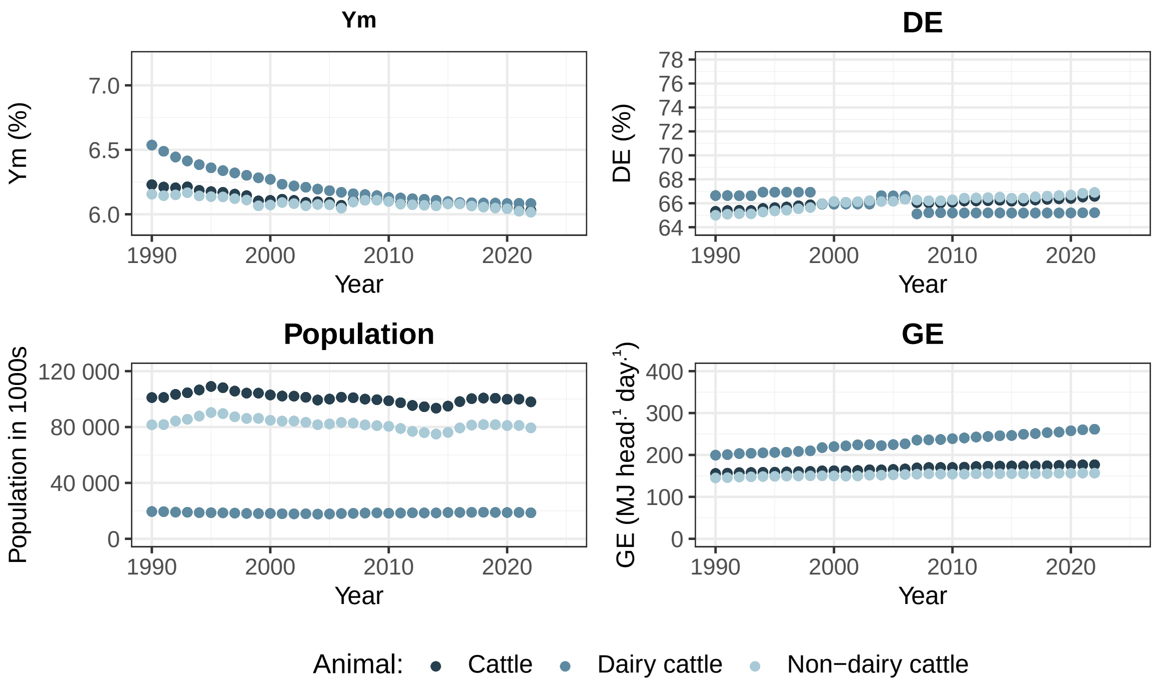

Figure A4: Time series of methane conversionfactor (Ym), digestibility of food (DE), population-size and gross energy intake (GE) of cattle and its subcategories dairy and non-dairy for United States of America.The data for dairy and non-dairy cattle is from the published CRT. The values for cattle is calculated as weighted mean of its subcategories, where the population-size is used as weight.

B Tables

Table B1: Counted cells for *Methanobrevibacter wolinii SH^T* (*M. wolinii*) with an OD (600 nm) of 0.387. Cells were counted with a phase-contrast microscope and a Buerk counting chamber at 400x magnification. Per chamber filling four square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean (ø cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	ø cellnumber
1	138	159	226	233	189
2	260	271	293	265	272.25
3	180	197	216	188	195.25
4	207	191	211	227	209

Table B2: Counted cells for *Methanobrevibacter wolinii SH^T* (*M. wolinii*) with an OD (600 nm) of 0.396. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean (ø cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	ø cellnumber
1	177	239	263	254	233.25
2	246	241	267	208	240.5
3	175	176	161	221	183.25
4	217	185	205	180	196.75

Table B3: Counted cells for *Methanobrevibacter thaueri CW^T* (*M. thaueri*) with an OD (600 nm) of 0.235. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean (ø cellnumber) per chamber filling was calculated. The culture was diluted 1:10 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	ø cellnumber
1	189	232	250	265	234
2	237	209	224	263	233.25
3	186	253	230	237	226.5
4	146	187	216	262	202.75

Table B4: Counted cells for *Methanobrevibacter thaueri CW*^{T} (*M. thaueri*) with an OD (600 nm) of 0.116. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean (ø cellnumber) per chamber filling was calculated. The culture was diluted 1:10 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	ø cellnumber
1	255	181	189	171	199
2	127	92	131	193	135.75
3	110	136	153	109	127
4	127	93	109	156	121.25

Table B5: Counted cells for *Methanobrevibacter millerae ZA-10*^{T} (*M. millerae*) with an OD (600 nm) of 0.378. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean (ø cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	Quadrat_5	ø cellnumber
1	186	175	173	156	220	182
2	118	204				161
3	211	178	232	211	184	203.2
4	128	128	201	204	160	164.2

Table B6: Counted cells for *Methanobrevibacter boviskoreani AbM4* (*M. boviskoreani*) with an OD(600 nm) of 0.287. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean (ø cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	ø cellnumber
1	123	162	166	170	155.25
2	162	228	205	242	209.25
3	170	128	137	213	162
4	187	174	263	237	215.25

Table B7: The total average of counted cells (ø cellnumber) for each *Methanobrevibacter* (*M.*) culture with corresponding OD at 600 nm (OD600). Counting was done with a phase-contrast microscope and a Buerk or Neubauer improved counting chamber at 400x magnification. ø cellnumber was calculated as arithmetic mean of shown cellcuonts in tab. B1 through B6. The dilution factor takes into account the dilution of the culture being counted.

culture	OD600	Dilution factor	ø cellnumber
M. wolinii	0.387	20	216.375
M. wolinii	0.396	20	213.438
M. thaueri	0.235	10	224.125
M. thaueri	0.116	10	145.750
M. millerae	0.378	20	177.600
M. boviskoreani	0.287	20	185.438

DECLARATION

I herewith declare that I have composed the present thesis myself and without use of any other than the cited sources and aids. Sentences or parts of sentences quoted literally are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned. The thesis in the same or similar form has not been submitted to any examination body and has not been published. This thesis was not yet, even in part, used in another examination or as a course performance. Furthermore, I declare that the submitted written (bound) copies of the present thesis and the version submitted on a data carrier are consistent with each other in contents.

Place, date

Signature

Table B8: Lists saved CH_4 for five scenarios of different success in shifting the microbiom from methanogen towards non-methanogen (10, 20, 30, 40 and 50%). It was calculated an average for the decade 2010 to 2019 for the three countries Germany (DEU), New Zealand (NZL), United States of America (USA). The CO_2 eq. was calculated for the GWP20 and GWP100 with factors taken from **?**, Table 7.15, Chapter 7 (79.7 for GWP20 and 27 for GWP100)

country	red.-scenario	CH_4 [kt]	GWP20 [kt CO_2 eq.]	GWP100 [kt CO_2 eq.]
DEU	10	104.40	8 320.60	2 818.77
DEU	20	206.52	16 459.82	5 576.10
DEU	30	306.45	24 424.23	8 274.20
DEU	40	404.27	32 220.05	10 915.20
DEU	50	500.04	39 853.21	13 501.09
NZL	10	88.23	7 031.61	2 382.10
NZL	20	174.47	13 905.60	4 710.81
NZL	30	258.82	20 627.83	6 988.10
NZL	40	341.33	27 203.88	9 215.87
NZL	50	422.07	33 639.03	11 395.91
USA	10	749.55	59 738.75	20 237.72
USA	20	1 479.97	117 953.24	39 959.07
USA	30	2 192.06	174 707.16	59 185.61
USA	40	2 886.58	230 060.64	77 937.73
USA	50	3 564.25	284 070.52	96 234.68

Table B9: Listed linear models for predicting relevant values for calculating 2030 CH_4 emission of **Germany**. The relevant indicators are total methane emission without cattle enteric fermentation (EnterFe) and Ym (methane conversion factor), DE (digestibility), NE_x (net energy intake), population size for dairy cattle (DC) and non-dairy cattle (NDC). For fitting the linear models three different time periods were used (the last 5, 10 or 15 years of the data set). The fitted value is thereby depended on the time. Besides the equation, the R^2 is given as well. R^2 for the 15-year model of "NDC YM [%]" turns zero because no change is predicted.

time sample	fitted value	equation	R^2
5	total CH_4 without cattle EnterFe [kt]	$y = -6.25 \cdot x + 13677.282$	0.070
5	DC population size [1000s]	$y = -78.386 \cdot x + 162269.508$	0.956
5	DC Ym [%]	$y = -0.023 \cdot x + 52.062$	0.712
5	DC DE [%]	$y = 0.063 \cdot x - 51.879$	0.735
5	DC NE_x [MJ head^{-1} day^{-1}]	$y = 1.868 \cdot x - 3630.669$	0.720
5	NDC population size [1000s]	$y = -110.076 \cdot x + 229727.445$	0.893
5	NDC Ym [%]	$y = 0.005 \cdot x - 4.499$	0.821
5	NDC DE [%]	$y = -0.035 \cdot x + 144.304$	0.891
5	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.129 \cdot x - 215.365$	0.888
10	total CH_4 without cattle EnterFe [kt]	$y = -36.946 \cdot x + 75747.569$	0.792
10	DC population size [1000s]	$y = -76.559 \cdot x + 158578.922$	0.988
10	DC Ym [%]	$y = -0.024 \cdot x + 55.219$	0.946
10	DC DE [%]	$y = 0.057 \cdot x - 40.026$	0.932
10	DC NE_x [MJ head^{-1} day^{-1}]	$y = 1.941 \cdot x - 3778.256$	0.946
10	NDC population size [1000s]	$y = -168.698 \cdot x + 348278.384$	0.974
10	NDC Ym [%]	$y = 0.002 \cdot x + 2.939$	0.483
10	NDC DE [%]	$y = -0.016 \cdot x + 106.264$	0.749
10	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.209 \cdot x - 377.797$	0.957
15	total CH_4 without cattle EnterFe [kt]	$y = -40.092 \cdot x + 82103.312$	0.934
15	DC population size [1000s]	$y = -44.459 \cdot x + 93727.349$	0.767
15	DC Ym [%]	$y = -0.023 \cdot x + 52.942$	0.977
15	DC DE [%]	$y = 0.032 \cdot x + 10.375$	0.716
15	DC NE_x [MJ head^{-1} day^{-1}]	$y = 1.532 \cdot x - 2951.872$	0.940
15	NDC population size [1000s]	$y = -125.537 \cdot x + 261079.881$	0.917
15	NDC Ym [%]	$y = 0 \cdot x + 6.57$	0.000
15	NDC DE [%]	$y = -0.014 \cdot x + 102.431$	0.852
15	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.144 \cdot x - 246.555$	0.863

Table B10: Listed linear models for predicting relevant values for calculating 2030 CH_4 emission of **New Zealand**. The relevant indicators are total methane emission without cattle enteric fermentation (EnterFe) and Ym (methane conversion factor), DE (digestibility), NE_x (net energy intake), population size for dairy cattle (DC) and non-dairy cattle (NDC). For fitting the linear models three different time periods were used (the last 5, 10 or 15 years of the data set). The fitted value is thereby depended on the time. Besides the equation, the R^2 is given as well. R^2 is NA for indicators where the variance of the values is zero. In addition there are two zero values because of rounding, first the slope of the 15-year model for "NDC DE [%]" and the R^2 for the 15-year model of "NDC population size [1000s]".

time sample	fitted value	equation	R^2
5	total CH_4 without cattle EnterFe [kt]	$y = -12.753 \cdot x + 26300.235$	0.997
5	DC population size [1000s]	$y = -102.756 \cdot x + 213780.671$	0.887
5	DC Ym [%]	$y = 0 \cdot x + 6.5$	NA
5	DC DE [%]	$y = 0.011 \cdot x + 54.263$	0.570
5	DC NE_x [MJ head^{-1} day^{-1}]	$y = 0.715 \cdot x - 1352.955$	0.797
5	NDC population size [1000s]	$y = -71.703 \cdot x + 148784.07$	0.732
5	NDC Ym [%]	$y = 0 \cdot x + 6.5$	NA
5	NDC DE [%]	$y = 0 \cdot x + 70.346$	NA
5	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.937 \cdot x - 1835.457$	0.852
10	total CH_4 without cattle EnterFe [kt]	$y = -10.058 \cdot x + 20851.018$	0.978
10	DC population size [1000s]	$y = -88.716 \cdot x + 185394.664$	0.930
10	DC Ym [%]	$y = 0 \cdot x + 6.5$	NA
10	DC DE [%]	$y = -0.007 \cdot x + 90.206$	0.305
10	DC NE_x [MJ head^{-1} day^{-1}]	$y = 0.906 \cdot x - 1740.452$	0.895
10	NDC population size [1000s]	$y = 22.947 \cdot x - 42611.147$	0.211
10	NDC Ym [%]	$y = 0 \cdot x + 6.5$	NA
10	NDC DE [%]	$y = 0 \cdot x + 70.346$	NA
10	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.508 \cdot x - 968.768$	0.773
15	total CH_4 without cattle EnterFe [kt]	$y = -9.839 \cdot x + 20408.364$	0.992
15	DC population size [1000s]	$y = -29.407 \cdot x + 65583.652$	0.219
15	DC Ym [%]	$y = 0 \cdot x + 6.5$	NA
15	DC DE [%]	$y = 0.021 \cdot x + 33.254$	0.469
15	DC NE_x [MJ head^{-1} day^{-1}]	$y = 1.019 \cdot x - 1969.163$	0.966
15	NDC population size [1000s]	$y = -0.097 \cdot x + 3942.39$	0.000
15	NDC Ym [%]	$y = 0 \cdot x + 6.5$	NA
15	NDC DE [%]	$y = 0 \cdot x + 71.161$	0.481
15	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.105 \cdot x - 153.685$	0.096

Table B11: Listed linear models for predicting relevant values for calculating 2030 CH_4 emission of the **USA**. The relevant indicators are total methane emission without cattle enteric fermentation (EnterFe) and Ym (methane conversion factor), DE (digestibility), NE_x (net energy intake), population size for dairy cattle (DC) and non-dairy cattle (NDC). For fitting the linear models three different time periods were used (the last 5, 10 or 15 years of the data set). The fitted value is thereby depended on the time. Besides the equation, the R^2 is given as well. Slope of the 15-year model for "DC DE [%]" is zero due to rounding issues.

time sample	fitted value	equation	R^2
5	total CH_4 without cattle EnterFe [kt]	$y = -529.671 \cdot x + 1091556.297$	0.989
5	DC population size [1000s]	$y = -78.221 \cdot x + 176838.967$	0.789
5	DC Ym [%]	$y = -0.001 \cdot x + 8.017$	0.704
5	DC DE [%]	$y = 0.006 \cdot x + 53.297$	0.845
5	DC NE_x [MJ head^{-1} day^{-1}]	$y = 0.736 \cdot x - 1400.748$	0.977
5	NDC population size [1000s]	$y = -526.204 \cdot x + 1143925.596$	0.765
5	NDC Ym [%]	$y = -0.01 \cdot x + 27.096$	0.960
5	NDC DE [%]	$y = 0.081 \cdot x - 96.049$	0.952
5	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.173 \cdot x - 295.476$	0.921
10	total CH_4 without cattle EnterFe [kt]	$y = -245.886 \cdot x + 518212.392$	0.753
10	DC population size [1000s]	$y = 21.479 \cdot x - 24555.96$	0.144
10	DC Ym [%]	$y = -0.003 \cdot x + 12.927$	0.802
10	DC DE [%]	$y = 0.002 \cdot x + 62.028$	0.293
10	DC NE_x [MJ head^{-1} day^{-1}]	$y = 0.671 \cdot x - 1269.793$	0.990
10	NDC population size [1000s]	$y = 638.685 \cdot x - 1209266.975$	0.544
10	NDC Ym [%]	$y = -0.006 \cdot x + 19.106$	0.787
10	NDC DE [%]	$y = 0.049 \cdot x - 32.878$	0.820
10	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.113 \cdot x - 173.444$	0.838
15	total CH_4 without cattle EnterFe [kt]	$y = -236.704 \cdot x + 499649.489$	0.876
15	DC population size [1000s]	$y = 35.307 \cdot x - 52471.673$	0.536
15	DC Ym [%]	$y = -0.005 \cdot x + 16.261$	0.916
15	DC DE [%]	$y = 0 \cdot x + 65.618$	0.011
15	DC NE_x [MJ head^{-1} day^{-1}]	$y = 0.617 \cdot x - 1159.782$	0.993
15	NDC population size [1000s]	$y = 104.563 \cdot x - 131253.734$	0.040
15	NDC Ym [%]	$y = -0.006 \cdot x + 17.665$	0.876
15	NDC DE [%]	$y = 0.043 \cdot x - 19.257$	0.890
15	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.106 \cdot x - 158.969$	0.915